

grok-wsl-ubuntu / 019f451f-06c1-7691-98f4-9111a1f767b7

Metadata

```
- Source: `grok-wsl-ubuntu`
- Kind: `grok`
- Source file:
`\\wsl.localhost\Ubuntu\home\avidullu\grok\sessions\%2Fhome%2Favidullu%2Fprojects%2Fkhelsutra-guru%2Fbadminton-highlight-indexer\019f451f-06c1-7691-98f4-9111a1f767b7\chat_history.jsonl`
- SHA-256: `0b7832dc49f2eab95157861445c27b090f22f88628973908319f9b287f25e4b1`
- Source modified: `2026-07-09T05:13:05+00:00`
- Imported at: `2026-07-09T08:32:30+00:00`
- project: `%2Fhome%2Favidullu%2Fprojects%2Fkhelsutra-guru%2Fbadminton-highlight-indexer`
- session_id: `019f451f-06c1-7691-98f4-9111a1f767b7`
```

Transcript

1. system

You are Grok 4.5 released by xAI. You are an interactive CLI tool that helps users with software engineering tasks. Your main goal is to complete the user's request, denoted within the `<user_query>` tag.

`<executing_actions_with_care>`

Carefully consider the reversibility and blast radius of actions. Generally you can freely take local, reversible actions like editing files or running tests. But for actions that are hard to reverse, affect shared systems beyond your local environment, or could otherwise be risky or destructive, check with the user before proceeding. The cost of pausing to confirm is low, while the cost of an unwanted action (lost work, unintended messages sent, deleted branches) can be very high. For actions like these, consider the context, the action, and user instructions, and by default transparently communicate the action and ask for confirmation before proceeding. This default can be changed by user instructions - if explicitly asked to operate more autonomously, then you may proceed without confirmation, but still attend to the risks and consequences when taking actions. A user approving an action (like a git push) once does NOT mean that they approve it in all contexts, so unless actions are authorized in advance in durable instructions like AGENTS.md files, always confirm first. Authorization stands for the scope specified, not beyond. Match the scope of your actions to what was actually requested.

Examples of the kind of risky actions that warrant user confirmation:

- Destructive operations: deleting files/branches, dropping database tables, killing processes, `rm -rf`, overwriting uncommitted changes
- Hard-to-reverse operations: force-pushing (can also overwrite upstream), `git reset --hard`, amending published commits, removing or downgrading packages/dependencies, modifying CI/CD pipelines
- Actions visible to others or that affect shared state: pushing code, creating/closing/commenting on PRs or issues, sending messages (Slack, email, GitHub), posting to external services, modifying shared infrastructure or permissions
- Uploading content to third-party web tools (diagram renderers, pastebins, gists) publishes it
- consider whether it could be sensitive before sending, since it may be cached or indexed even if later deleted.

When you encounter an obstacle, do not use destructive actions as a shortcut to simply make it go away. For instance, try to identify root causes and fix underlying issues rather than bypassing safety checks (e.g. `--no-verify`). If you discover unexpected state like unfamiliar files, branches, or configuration, investigate before deleting or overwriting, as it may represent the user's in-progress work. For example, typically resolve merge conflicts rather than discarding changes; similarly, if a lock file exists, investigate what process holds it rather than deleting it. In short: only take risky actions carefully, and when in doubt, ask before acting. Follow both the spirit and letter of these instructions - measure twice, cut once.

`</executing_actions_with_care>`

`<tool_calling>`

- Use specialized tools instead of bash commands when possible, as this provides a better user experience. For file operations, prefer dedicated file tools (e.g., `read_file` for reading files instead of `cat/head/tail`, `search_replace` for editing and creating files instead of `sed/awk`). Reserve bash tools exclusively for actual system commands and terminal operations that require shell execution. NEVER use bash `echo` or other command-line tools to communicate thoughts, explanations, or instructions to the user. Output all communication directly in your response text instead.

</tool_calling>

<background_tasks>

For watch processes, polling, and ongoing observation (CI status, log tailing, API polling): Use the `monitor` tool ? it streams each stdout line back as a chat notification.

</background_tasks>

<output_efficiency>

- Write like an excellent technical blog post ? precise, well-structured, and clear, in complete sentences. Most responses should be concise and to the point, but the quality of prose should be high.

- Same standards for commit and PR descriptions: complete sentences, good grammar, and only relevant detail.

- Prefer simple, accessible language over dense technical jargon. Explain what changed and why in plain language rather than listing identifiers. Stay focused: avoid filler, repetition, over-the-top detail, and tangents the user did not ask for.

- Keep final responses proportional to task complexity.

</output_efficiency>

<formatting>

Your text output is rendered as GitHub-flavored markdown (CommonMark). Use markdown actively when it aids the reader: bullet lists for parallel items, **bold** for emphasis, `inline code` for identifiers/paths/commands, and tables for short enumerable facts (file/line/status, before/after, quantitative data). Don't pack explanatory reasoning into table cells ? explain before or after the table. Match structure to the task: a simple question gets a direct answer in prose, not headers and numbered sections.

</formatting>

<user_guide>

Documentation about the Grok Build TUI ? including configuration, keyboard shortcuts, MCP servers, skills, theming, plugins, and more ? is stored as `.md` files in `~/grok/docs/user-guide/`. When users ask about features or how to use the TUI, read the relevant file from that directory. Present the information directly.

</user_guide>

2. user

<user_info>

OS Version: linux

Shell: /bin/bash

Workspace Path: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer

Today's date: 2026-07-09

Note: Prefer using relative paths over absolute paths as tool call args when possible.

</user_info>

<git_status>

This is the git status at the start of the conversation. Note that this status is a snapshot in time, and will not update during the conversation.

On branch fix/534-l4-detector-progress-telemetry

Your branch is up to date with 'origin/fix/534-l4-detector-progress-telemetry'.

nothing to commit, working tree clean

</git_status>

3. user

<system-reminder>

As you answer the user's questions, you can use the following context (ordered from repo root to current directory - deeper files take precedence on conflicts):

From: /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/CLAUDE.md CLAUDE.md ? project memory / agent handoff

Local, AI-powered **badminton** (? racket-sports) highlight indexer + rally detector:
FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
repos: `sports-data-collector`, a VLC-based `rally-annotator`.

Read these first

- **docs/README.md** ? doc index. **docs/CODE_MAP.md** ? code navigation + data contracts (the cold-start map; read before touching code).
- **Authoritative owned-model roadmap:** **docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md** (+ the live **docs/NEXT_STEPS.md**). These SUPERSEDE the 2026-06-07 strategy docs below where they disagree (licensing, Gemma-4 stance, base model, conversational-QA).
- **Strategy (post-scaffolding):** **docs/COMMERCIALIZATION.md**, **docs/PMF_MARKET_RESEARCH.md**, **docs/PLATFORM_ARCHITECTURE.md**, **docs/DEFERRED_HARDENING_PLAN.md**. Also **docs/ROADMAP.md** (offline-segmenter narrative) and **docs/BACKLOG.md**.
- **Rally-detection quality track** (design merged 2026-06-13, owner-gated, implementation in progress):
 - docs/HOW_RALLY_DETECTION_WORKS.md** (2-layer mental model) ?
 - docs/MULTI_SIGNAL_FUSION_PLAN.md** (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built **rally_gate.py** "Gate A" ? implemented in **backend/eval/served_gate_a.py**, default-OFF) ?
 - docs/EXPERIMENT_HARNESS.md** (A/B + shadow eval so cues land flag-gated/default-OFF with zero regression) ?
 - docs/ROORA_LABELING_GUIDELINES.md** (v8 vendor labeling spec). **docs/NEXT_STEPS.md** has the prioritized pick-up for this track.
- **History:** **docs/archives/** ? ADRs (**decisions/DECISIONS.md**), research, and **checkpoints/** (latest: **2026-06-strategy-foundation/**). **Archive policy:** only move **terminal-state** (DONE/REJECTED) work; live docs stay at **docs/** top level ? and **before** archiving, HONESTLY update the doc first (verify claims vs code ? flip status + a completion note ? update inbound refs ? keep the lineage ? then **git mv**), per the checklist + living lifecycle register in **docs/DOC_STATUS.md**.

Current state (July 2026)

- **Platform slice SHIPPED:** async job model (**backend/api/job_worker.py**, #339), **StorageBackend** + **ComputeBackend** registries with concrete implementations (**backend/storage/** ? local/gcs/gdrive/gdrive_api/s3; **backend/compute/** ? local/cloud_run), capabilities/assets endpoints, and the data flywheel (**backend/flywheel.py**). **COMPUTE_DECOUPLED_SERVING** M0?M9 largely shipped (ADR-014 ratified 2026-06-21); Gen-0 owned-model weights pushed (**weights/0.1.0-l4/**).
- **Launch decision:** NO-GO (owner, 2026-06-30, #404) ? the rally-detection product is NOT cleared for external/commercial launch; we stay in development. Two unblockers gate a future GO: (1) set + meet the ship-gate target (#172, needs corpus growth); (2) resolve WASB shuttle-detector weight provenance (commercial-clean licensing).
- **Active tracks** (see **docs/NEXT_STEPS.md** for the prioritized pending list): rally-quality / owned-model ML work (gated on golden-corpus growth), packaging (**docs/PACKAGING_PLAN.md**), and the khelsutra SPA/deploy sibling track.

Architecture in one breath

- **Pluggable registries** (ABC + **Registry** singleton): segmenters / providers / validators / sports / annotations ? see **backend/pipeline/segmenters/base.py**. **StorageBackend** + **ComputeBackend** registries are **implemented** (**backend/storage/**, **backend/compute/**; **PLATFORM** §3 ? realized by **COMPUTE_DECOUPLED_SERVING**).
- **video_id** = MD5 of the file ? **backend/utils/hashing.py::compute_video_id** (one helper).
- Typed config = **backend/config/** (Pydantic). DB schema evolves via the **PRAGMA user_version** migration ladder in **backend/infrastructure/database.py**.

Committed guardrails (non-negotiable)

- ****Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a training data source**** (terms/copyright). Commercial-clean licensing for EVERY dependency ? code, data, AND weights: ****MIT / Apache-2.0 / BSD only**** (ratified 2026-06-09, owner-agreed).
- ****Owned-model base:**** ****Qwen3-VL (Apache-2.0)**** primary, InternVL3 (MIT) fallback; ****Gemma-4 is AMBER / bench-only**** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; do NOT build on it without sign-off). ****Reject Gemma 3**** (viral license). ****Exclude minors**** from the training pool; per-user training data needs a separate explicit opt-in.
- ****v1 = rally + highlight + history + correction loop****; ****gait/biometrics strictly future**** (GDPR Art. 9).
- ****Generic data schema****; ``coach ? athlete`` is an ****optional overlay****, not baked in.
- ****Compute = GCP ONLY (ADR-013, 2026-06-20):**** GPU training + serving run on ****GCP**** (``deploy/gcp/``;
Cloud Run+GPU for serving per ADR-012). ****DigitalOcean is DROPPED for compute**** (DO GPU not enabled
on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't re-evaluate
DO/Paperspace for compute without a new ADR; the ``$200`` DO credits go to ****non-compute only****.
The
``deploy/digitalocean/`` scripts are kept solely as provider-agnostic Linux-GPU setup reused by GCP.

Workflow conventions

- ****Branch off `master`**; ONE PR per work item; NO PR stacking. **** Don't open a PR unless asked.**
- Tests: ``python run_all_tests.py`` (offline, fully mocked). Python 3.10+ (the ``trackers`` package requires ?3.10); ``ffmpeg`` required.
- Never push to the default branch directly; descriptive commit messages.
- ****Substantial design/project work gets a tracked project doc**** following ``docs/PROJECT_DOC_TEMPLATE.md`` (status header · decisions-locked · progress tracker · definition-of-done; archived on completion). Precedents: ``docs/RALLY_QUALITY_RESEARCH.md`` §7, ``docs/GOLDEN_REGRESSION_FIXTURES.md``.
- ****Launch / "bit-flip" PRs**** (a default-flip that drives a tracked project to terminal state) MUST, in the SAME PR, run the ``docs/DOC_STATUS.md`` §2 archival due-diligence on that project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR lineage ? ``git mv`` to ``docs/archives/`` ? record in DOC_STATUS §3), and fix any internal-link breakage the move causes. Enforced by ``.github/PULL_REQUEST_TEMPLATE.md`` + the ``link-check`` CI job (``python -m tools.check_md_links``).
- ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to the owner,
who has the golden corpus + labelled videos; verify pure-logic paths locally.

Follow these instructions exactly. When working in subdirectories not listed above, check for additional project instruction files (AGENTS.md, Claude.md, etc.).

</system-reminder>

4. user

<system-reminder>

The following skills are available for use:

- **check-work**: Check your work with a verification subagent that reviews diffs, runs builds and tests, and evaluates correctness. Read this file for instructions
Use when: asked to "check work", "verify changes", "self-verify", "/check-work", "/check", "/verify", or "/self-verify".
Absolute path: `/home/avidullu/.grok/skills/check-work/SKILL.md`
- **pptx**: Use this skill any time a .pptx file is involved in any way ? as input, output, or both. This includes: creating slide decks, pitch decks, or presentations; reading, parsing, or extracting text from any .pptx file (even if the extracted content will be used elsewhere, like in an email or summary); editing, modifying, or updating existing presentations; combining or splitting slide files; work?
Absolute path: `/home/avidullu/.grok/skills/pptx/SKILL.md`
- **create-skill**: Interactively create a new Grok skill (SKILL.md + optional scripts/references)
Use when: the user wants to create a skill, scaffold a skill, or runs `/create-skill`.

Absolute path: /home/avidullu/.grok/skills/create-skill/SKILL.md

- xlsx: Use this skill any time a spreadsheet file is the primary input or output. This means any task where the user wants to: open, read, edit, or fix an existing .xlsx, .xlsm, .csv, or .tsv file (e.g., adding columns, computing formulas, formatting, charting, cleaning messy data); create a new spreadsheet from scratch or from other data sources; or convert between tabular file formats. Trigger espec?

Absolute path: /home/avidullu/.grok/skills/xlsx/SKILL.md

- help: Grok documentation and configuration help

Use when: users ask about setup, configuration, MCP servers, authentication, skills, slash commands, keyboard shortcuts, or any Grok feature. Also use proactively when you detect a user is having trouble with setup or onboarding.

Absolute path: /home/avidullu/.grok/skills/help/SKILL.md

- imagine: How to use the image_gen and image_edit tool calls in Grok Build: when to build a visual with code instead of generating it, prompt-craft, reference-first handling of real people, factual grounding, and asset-consistency. Load this whenever generating or editing an image is on the table, i.e. when an image_gen or image_edit call is being considered or about to be made. Tool-usage-driven, not tr?

Absolute path: /home/avidullu/.grok/skills/imagine/SKILL.md

- docx: Use this skill whenever the user wants to create, read, edit, or manipulate Word documents (.docx files). Triggers include: any mention of 'Word doc', 'word document', '.docx', or requests to produce professional documents with formatting like tables of contents, headings, page numbers, or letterheads. Also use when extracting or reorganizing content from .docx files, inserting or replacing ima?

Absolute path: /home/avidullu/.grok/skills/docx/SKILL.md

- review: Run a reviewer subagent against uncommitted local changes, a named branch, or a GitHub PR. Local and branch modes write a review file plus a summary to disk. PR mode posts the findings as a PENDING GitHub review for the user to inspect and submit through the UI.

Use when: asked to 'review', 'code review', 'review my changes', 'review this PR', or '/review'.

Absolute path: /home/avidullu/.grok/bundled/skills/review/SKILL.md

- pr-babysit: Monitor PRs, fix CI failures, address review comments, resolve merge conflicts, and restack stacks. Supports independent PRs, Graphite stacks, and GitHub stacked PRs (gh-stack).

Use when: "/pr-babysit".

Absolute path: /home/avidullu/.grok/bundled/skills/pr-babysit/SKILL.md

- design: Run the full design-doc-writer and design-doc-reviewer loop until consensus. Produces a polished design document with a PR plan.

Use when: asked to "design", "write a design doc", "system design", "architecture doc", "technical spec", or "/design".

Absolute path: /home/avidullu/.grok/bundled/skills/design/SKILL.md

- execute-plan: Execute a PR Plan DAG from a design document. Parses the plan, topologically sorts it, implements PRs in parallel using worktree-isolated subagents, runs mandatory orchestrator-level review, and assembles either a Graphite PR stack or a plain-git branch stack depending on tool availability.

Use when: asked to "execute plan", "run the plan", "implement the design", or "/execute-plan".

Absolute path: /home/avidullu/.grok/bundled/skills/execute-plan/SKILL.md

</system-reminder>

5. user

<system-reminder>

MCP servers connected:

- github (91 tools)
- gmail (16 tools)
- google_calendar (8 tools)
- google_drive (5 tools)
- outlook (18 tools)
- tasks (6 tools)

To use MCP tools, you MUST call `search\_tool` first to retrieve the tool's input schema before calling `use\_tool`. NEVER guess parameter names ? always use the exact schema returned by `search\_tool`.

</system-reminder>

6. user

```
<user_query>
Hello. Can you ramp up and review https://github.com/Khelsutra/badminton-highlight-
indexer/pull/547 and leave due comments on the PR for the author to address?
</user_query>
```

7. assistant

I'll review PR #547 and leave comments for the author. Loading the review skill and gathering the PR context.

8. tool_result

```
1?---
name: review
description: >-
  Run a reviewer subagent against uncommitted local changes, a named branch,
  or a GitHub PR. Local and branch modes write a review file plus a summary to disk.
  PR mode posts the findings as a PENDING GitHub review for the user to inspect and
  submit through the UI.
when-to-use: "Use when asked to 'review', 'code review', 'review my changes', 'review this PR',
or '/review'."
argument-hint: "[--local | --branch <name> | --pr <number-or-url> | <auto-detect>]"
10?---
```

Review Skill

You are an orchestrator that runs a reviewer subagent against one of three review targets. You coordinate only ? **all** review findings are authored by a subagent whose prompt is seeded with the `reviewer` persona instructions, never by the orchestrator directly.

Persona Injection

This skill uses the **reviewer** persona. The persona instructions are defined at:

```
20?```
<dirname of this SKILL.md>/../shared/personas/reviewer.md
```
```

Resolve this path once at the start of the run (the system context gives you the absolute path to this SKILL.md). Read the file with `read\_file` and store its contents as `reviewer\_persona\_instructions`.

When launching the reviewer subagent, **prepend** the persona instructions to the prompt. Do NOT pass a `persona` parameter to `spawn\_subagent` ? that parameter is not supported. Instead, prefix the `description` with `[reviewer]` so the pager's subagent label renderer surfaces "Reviewer" at the top of the subagent row (see Step 2 below).

1. **Local mode (default)** -- uncommitted local changes (staged + unstaged + untracked).
2. **Branch mode** -- the diff between a named branch and its merge-base with the default base branch.
- 30?3. **PR mode** -- a GitHub pull request. Findings are posted as a PENDING review for the user to inspect and submit through GitHub.

The reviewer subagent is read-only -- it never modifies code. The orchestrator never edits source either; the only artifacts produced are a review file, a summary file, and (in PR mode) a pending GitHub review.

## Invocation

The user runs:

```
```
/review                                # local mode (default)
```

```

40?/review --local                # local mode (explicit)
/review --branch <name>           # branch mode (explicit)
/review --pr <number-or-url>      # PR mode (explicit)
/review <plain-arg>              # auto-detect; see disambiguation below
...

```

Argument parsing

Parse the argument string with these deterministic rules, applied in order. The first rule that matches wins; do not fall through.

- 50?1. ****Empty / whitespace-only****: `MODE=local`, no target.
2. ****Starts with `--local`****: `MODE=local`. Reject any extra positional argument with an error.
3. ****Starts with `--branch <name>`****: `MODE=branch`, `TARGET=<name>`. The branch name is required -- if the flag appears with no following token (or only with another `--`-prefixed token), reject with `Flag --branch requires an argument: <branch-name>` and stop.
4. ****Starts with `--pr <id-or-url>`****: `MODE=pr`, `TARGET=<id-or-url>`. The id or URL is required -- if the flag appears with no following token (or only with another `--`-prefixed token), reject with `Flag --pr requires an argument: <number-or-url>` and stop.
5. ****Starts with `--` but does not match any of the above****: reject with `Unknown flag: <flag>`. Valid flags: `--local`, `--branch <name>`, `--pr <number-or-url>.` and stop. Do NOT fall through to auto-detect.
6. ****Plain argument given (no flag prefix)****: auto-detect against the rules below, also applied in order:
 1. Matches the regex `^https?://github\.com/[^\s]+/[^\s]+/pull/\d+(?:[/?#].\*)?\$` -- treat as PR URL. `MODE=pr`, `TARGET=<url>`.
 2. Matches `^#\d+\$` (optional leading `#`, then pure digits) -- treat as PR number. `MODE=pr`, `TARGET=<digits without leading #>`.
 3. Resolves to a local or remote branch via `git rev-parse --verify --quiet <arg>` or `git rev-parse --verify --quiet origin/<arg>` -- treat as branch. `MODE=branch`, `TARGET=<arg>` (use the bare name, not `origin/<arg>`).
 4. None of the above -- ask the user whether the argument is a PR identifier or a branch name (use the appropriate ask/question tool if available). Provide three options: "PR (treat as PR identifier)", "Branch (treat as branch name)", and "Cancel". On Cancel, stop.

60?

If the user passes both a flag and a positional argument (e.g., `/review --local somebranch`), reject with a clear error message and stop. The flags are mutually exclusive.

Setup

Generate a unique ID for this run's artifact files. Execute this via `run\_terminal\_cmd` and capture stdout:

```

```bash
python3 -c "import uuid; print(uuid.uuid4().hex[:8])"
```

```

70?

This matches the pattern used by `implement/SKILL.md`. The previous draft of this skill chained two fallbacks (`/proc/sys/kernel/random/uuid` and `date +%s`), but the `/proc` path is missing on macOS and the `date` fallback's `tail -c 9` kept a trailing newline -- both bugs. `python3` is reliably present in the supported environments; if it is genuinely absent, the validation step below catches it with a clear error.

****Validate**** that the command produced a non-empty 8-character string. If `REVIEW\_ID` is empty or the command failed, report the error to the user (with the suggestion to install Python 3) and stop -- do not proceed with empty/malformed file paths.

Store the output as `REVIEW\_ID`. Set a restrictive umask first so all subsequent artifact writes land at mode 0600 -- the diff and review files can capture `.env` snippets or other secrets and the default 0644 leaks them to other users on shared hosts:

```

```bash
umask 077
```

```

80?

Then define the file paths used throughout the run:

```
- `summary_file`: `/tmp/grok-review-summary-`${REVIEW_ID}.md` (orchestrator-written; used in local and branch modes only -- not written or read in PR mode)
- `review_file`: `/tmp/grok-review-`${REVIEW_ID}.md` (reviewer subagent writes here; produced in all modes)
- `diff_file`: `/tmp/grok-review-diff-`${REVIEW_ID}.diff` (collected diff fed to the reviewer; produced in all modes)
- `pending_review_payload`: `/tmp/grok-review-pending-`${REVIEW_ID}.json` (PR mode only -- not written or read in local/branch modes, so cleanup of this path is a no-op outside PR mode)
```

Initialize state variables:

```
90?- `mode`: one of `local`, `branch`, `pr` (set by argument parsing).
- `target`: the branch name, PR number, or PR URL (empty in local mode).
- `head_sha`, `base_sha`, `owner`, `repo`, `pr_number`, `pr_url`, `pr_title` (populated in PR mode by Step 1).
- `changed_files`: list of file paths in the diff (populated by Step 1).
```

Step 1: Resolve target & collect diff

The diff collection commands differ per mode.

Local mode

100?

1. Detect changes:

```
```bash
git status --porcelain
```
```

If the output is empty, print "No local changes to review (working tree clean)." Skip directly to Step 4 (Cleanup -- use the local/branch sub-case; both ``<diff_file>`` and the helper file list will be no-op `rm -f` since neither was written yet) and then Step 5 (Final report). The Final report should use the "Local / branch (empty-diff exit)" bullet. Do NOT launch the reviewer.

2. Build a unified diff covering staged + unstaged tracked changes AND untracked files:

110?

```
```bash
Staged + unstaged tracked changes (includes deletions and modifications).
Guard against fresh `git init` repos with no commits -- `git diff HEAD`
fails there with "ambiguous argument 'HEAD'". In that case, leave the
tracked-change portion empty and let the untracked loop populate the file.
`core.quotepath=false` keeps non-ASCII / space-bearing paths unquoted so
the parser in Step 3 PR mode sees literal paths.
if git rev-parse --verify --quiet HEAD >/dev/null; then
 git -c core.quotepath=false diff HEAD > "${diff_file}"
120? else
 : > "${diff_file}"
fi

Append each untracked file as an added-file diff (skip ignored files)
git ls-files --others --exclude-standard -z | while IFS= read -r -d '' f; do
 git -c core.quotepath=false diff --no-index -- /dev/null "$f" >> "${diff_file}" || true
done

Size check: print the byte count so the orchestrator can gate continuation.
130? # See the executable size check handling immediately below.
wc -c "${diff_file}"
```
```

Trade-off note: `git diff --no-index` exits with status 1 when differences are present (which

is always, here), so we suppress its exit with `|| true`. The alternative -- `git add -N <untracked> && git diff HEAD && git rm --cached <untracked>` -- mutates the index and risks leaving the user in an unexpected state if interrupted; the `--no-index` approach is non-mutating, which is why this block uses it.

****Fresh-repo guard****: on a brand-new `git init` with zero commits there is no `HEAD` to diff against, so the `if git rev-parse --verify --quiet HEAD` branch above falls through to the `else` and `\${diff\_file}` starts empty. Everything in the working tree at that point is untracked from git's perspective, so the subsequent `git ls-files --others` loop captures it correctly. The rest of the local-mode flow is unchanged.

****Size gate (orchestrator-side)****: read the byte count emitted by `wc -c` and act on it:

- ****> 10 MB****: abort with an error telling the user to add the offending paths to `.gitignore` (point them at `git status --porcelain` plus `du -sh` to find the worst offenders -- typical culprits are an untracked `node\_modules/`, `.cache/`, `target/`, or a stray dataset). Do NOT launch the reviewer; run cleanup and stop.
- 140? - ****> 1 MB****: ask the user to confirm before continuing (use the appropriate ask/question tool if available). On decline, run cleanup and stop. The reviewer subagent has a context limit and a multi-MB diff will saturate it.
- ****otherwise****: proceed silently.

Edge cases to be aware of (these do not break the workflow, but the user should be told if they apply):

- ****Very large untracked files**** (binary blobs, generated artifacts that were never `.gitignore`-d) get appended to `\${diff\_file}` in full. The size gate above handles this -- the prose here is just a pointer to the executable check above.
- ****Symlinks**** surfaced by `git ls-files --others` produce a single `Symbolic link` line in the diff rather than file content; the reviewer can still report on the symlink itself but cannot inspect its target.

3. Capture the changed-files list (same fresh-repo guard as in step 2 -- if there is no `HEAD`, skip the tracked-name listing and rely entirely on `git ls-files --others`):

```
```bash
150? {
 if git rev-parse --verify --quiet HEAD >/dev/null; then
 git -c core.quotepath=false diff --name-only HEAD
 fi
 git ls-files --others --exclude-standard
} | sort -u > /tmp/grok-review-files-${REVIEW_ID}.txt
```
```

Read this into `changed\_files`.

160?### Branch mode

1. Determine the base branch. Try in order:

```
```bash
if git rev-parse --verify --quiet origin/main >/dev/null; then
 BASE=origin/main
elif git rev-parse --verify --quiet origin/master >/dev/null; then
 BASE=origin/master
else
170? BASE=""
fi
```
```

Use an explicit `if/elif/else` (not two unconditional `&&` lines) so that `origin/master` does not overwrite `origin/main` when both exist (which happens during master-to-main migrations and on mirror repos). The `>/dev/null` redirect prevents the SHA emitted by `git rev-parse` from leaking into the orchestrator's captured output.

If `BASE` is empty (neither ref exists), ask the user which base ref to compare against (use the appropriate ask/question tool if available) (offer the local default branch name(s) you can

detect via `git symbolic-ref refs/remotes/origin/HEAD`, plus an "Other" option).

2. Verify the target branch exists:

```
180?  ```bash
      git rev-parse --verify --quiet "${target}" || git rev-parse --verify --quiet
      "origin/${target}"
      ```
```

If neither resolves, report the error and stop.

3. Compute the merge base and collect the diff (`core.quotePath=false` prevents C-style path quoting so the parser in Step 3 PR mode sees literal paths -- `gh pr diff` does not quote paths, so PR mode is unaffected):

```
```bash
MERGE_BASE=$(git merge-base "${BASE}" "${target}")
190?  git -c core.quotePath=false diff "${MERGE_BASE}".. "${target}" > "${diff_file}"
      git -c core.quotePath=false diff --name-only "${MERGE_BASE}".. "${target}" > /tmp/grok-review-
      files-${REVIEW_ID}.txt
      ```
```

4. **Empty-diff handling**: if `\${diff\_file}` is empty (or contains only whitespace), print "Branch `\${target}` has no changes vs `\${BASE}`." Skip directly to Step 4 (Cleanup -- use the local/branch sub-case) and then Step 5 (Final report). The Final report should use the "Local / branch (empty-diff exit)" bullet to note that the branch has no changes vs its base. Do NOT launch the reviewer.

Read `changed\_files` from the names file.

## PR mode

200?1. Verify `gh` authentication:

```
```bash
gh auth status
```
```

If this exits non-zero, warn the user that the PR cannot be fetched without `gh` auth, then stop with instructions to run `gh auth login`.

2. Fetch PR metadata in a single round trip (works for both numeric IDs and full URLs). Note that `gh pr view --json` does NOT expose a `baseRepository` field (verified: `gh pr view 1 --json baseRepository` returns `Unknown JSON field: "baseRepository"`). The available repo fields are `headRepository`, `headRepositoryOwner`, and `isCrossRepository`. The owner/repo for the upstream where the PR lives must therefore be derived from the `url` field instead. The `files` field is also requested here so we do not need a second round trip:

```
210? ```bash
 gh pr view "${target}" --json number,title,body,headRefOid,baseRefOid,headRefName,baseRefName
 ,url,headRepository,headRepositoryOwner,isCrossRepository,files \
 > /tmp/grok-review-prmeta-${REVIEW_ID}.json
      ```
```

Parse the JSON and populate:

```
- `pr_number` from `.number`
- `pr_title` from `.title`
- `pr_url` from `.url`
- `head_sha` from `.headRefOid`
```

```
220?  - `base_sha` from `.baseRefOid`
```

```
- `owner`, `repo` -- parse from `pr_url` using the regex
```

`\${https}://github\.(com|(?P<owner>[^\s]+)/(?P<repo>[^\s]+)/pull/\d+`. The URL always points at the upstream where the PR lives, even for cross-repo PRs (`isCrossRepository: true`), so this is the correct source for the review-posting endpoint.

```
- `changed_files` -- extract via `jq -r '.files[].path' /tmp/grok-review-
```

prmeta-`{REVIEW_ID}.json` > `/tmp/grok-review-files-{REVIEW_ID}.txt`, then read the file.

****Validate**** that all required fields are non-empty: ``pr_number``, ``head_sha``, ``base_sha``, ``owner``, ``repo``. If any is missing or empty (which can happen for very old PRs, PRs in unusual states, or partial responses), surface the parsed JSON to the user and stop -- do not proceed to build a payload with ``null`` fields.

3. Fetch the diff:

```
```bash
gh pr diff "{target}" > "{diff_file}"
230? ```
```

4. **\*\*Empty-diff handling\*\***: if ``{diff_file}`` is empty, print "PR `#{pr_number}` has no changes." Skip directly to Step 4 (Cleanup -- use the "PR mode, no reviewer output" sub-case, which removes `<diff_file>`, the prmeta JSON, the files list, and is a no-op on `<review_file>` / `<pending_review_payload>` because neither was written) and then Step 5 (Final report). The Final report should use the "PR (empty-diff exit)" bullet to note that the PR has no changes. Do NOT launch the reviewer.

After Step 1, report progress: "Collected diff for `<mode>` target `<summary>`. Launching reviewer..."

## Step 2: Launch reviewer subagent

Launch a single reviewer subagent by calling ``spawn_subagent``. Emit the ``spawn_subagent`` tool call before producing any "reviewer is starting" narration; the post-launch progress message ("Review complete. Processing findings...") belongs in a later assistant message after the tool result is in hand.

240? ``spawn_subagent`` parameters:

- ``subagent_type``: ``"general-purpose"``
- ``description``: ``"[reviewer] <mode> <target-summary>"`` (e.g., ``"[reviewer] pr #4221"`` or ``"[reviewer] branch feature/foo"`` or ``"[reviewer] local changes"``). The ``[reviewer]`` prefix is parsed by the pager's subagent label renderer (see ``format_subagent_label`` in ``xai-grok-pager``) so the subagent row shows "Reviewer" instead of the generic "General" fallback. The bracketed prefix is stripped from the displayed description.

Build the prompt with the mode-specific context. **\*\*Prepend the reviewer persona instructions\*\*** (loaded during setup) to the prompt. Use this template:

```
...
<reviewer_persona_instructions>
```

250?---

You are reviewing code changes. Mode: `<mode>`.

```
Target: <target-summary-line>
<if PR mode: PR URL: <pr_url>>
<if PR mode: head SHA: <head_sha>, base SHA: <base_sha>>
<if branch mode: base: <BASE>, merge-base: <MERGE_BASE>, head: <target>>
```

```
The unified diff is at: <diff_file>
260?The list of changed files is at: /tmp/grok-review-files-{REVIEW_ID}.txt
```

Read the diff first to understand the scope. The diff alone is often not enough context, so you should also ``read_file`` the source files referenced in the diff to understand call sites, types, and surrounding logic before flagging issues.

Write your structured findings to: `<review_file>`

Format:

270?## Summary

<2 to 4 sentence overall assessment of the changes -- what they do, whether they look correct, the dominant risk areas. This goes at the very top of the file, before any individual issues.>

## Issues

### Issue 1 -- Severity: bug

- File: path/to/file.ext:LINE
- 280?- Description: <what is wrong>
- Suggestion: <how to fix>
- Status: open

### Issue 2 -- Severity: suggestion

- File: path/to/file.ext:LINE
- Description: ...
- Suggestion: ...
- Status: open

290?Severity must be one of: bug, suggestion, nit. Each issue's Status field must be set to "open" (as shown in the example above).

<if PR mode, include this paragraph verbatim:>

IMPORTANT: For each issue, the File line MUST reference a single line number on the RIGHT side of the diff (the line number in the new/post-change file, not the pre-change file). If a finding spans a range, pick the most representative single line on the RIGHT side. This requirement is mandatory because the orchestrator will post these findings as inline comments on the GitHub PR, and the GitHub API rejects comments that do not target a line present in the diff.

<end if>

300?

If the diff is genuinely fine and you have no issues, write the Summary and an empty `## Issues` section (or omit the Issues section entirely). Do not invent issues to fill space.

...

Wait for the subagent to complete. If it fails, report the error to the user and stop.

After it completes, verify that `<review\_file>` exists and is non-empty. If it does not, report the error and stop.

310?Save the returned `subagent\_id` for the report. The reviewer is not resumed; this is a one-shot review.

Report progress: "Review complete. Processing findings..."

## Step 3: Handle output based on mode

The post-processing differs between local/branch mode (write summary to disk) and PR mode (post pending review to GitHub).

### Local / Branch mode

320?1. Read `<review\_file>` via `read\_file`.

2. Count issues by counting heading lines that match the regex `^### Issue \d+ -- Severity: (bug|suggestion|nit)\$` and bucket them by the captured severity. Issues whose heading does not match this pattern (typo'd severity, missing severity field, malformed heading) are malformed -- log a one-line warning to the user listing the heading line and treat them as uncounted. Do not attempt to re-parse the body for a `Severity:` field; the heading is the canonical source. The same regex governs the parsing in Step 3 PR mode below.

3. Compute diff stats:

```
```bash
git diff --shortstat ...      # local: HEAD; branch: MERGE_BASE..target
```
```

For local mode, run `git diff --shortstat HEAD` and also count untracked files separately. For branch mode, run `git diff --shortstat "\${MERGE\_BASE}".. "\${target}"`.

330?4. Use the `write` tool to create `` with this structure:

```
```markdown
# Review Summary

- **Mode**: <mode>
- **Target**: <target-summary>
- **Files reviewed**: <count> (<list, truncated to 10 if longer>)
- **Diff stats**: <shortstat-line>
- **Issue counts**: <X> bugs, <Y> suggestions, <Z> nits

340?
## Top issues

<First 5 issue titles, one per line, in the form "[severity] file:line -- description
(truncated to ~100 chars)">

See the full review at: <review_file>
```
```

5. Print to the user:

- Inline issue counts and the file paths to `` and ``.

350?

Do NOT delete `` or `` in local/branch modes -- those files ARE the deliverable.

## PR mode

1. Read `` via `read\_file` and parse it into a structured representation:

- Extract the `## Summary` section -- everything after the `## Summary` heading and before the next `##` heading.

- Extract each issue by matching heading lines against `^### Issue \d+ -- Severity: (bug|suggestion|nit)\$` (same regex as Step 3 Local/Branch mode). For each matched block, capture:

- `severity` -- the captured group from the heading regex
- `file` and `line` -- parsed from the `File: path:line` field. If `:line` is missing, the issue cannot become an inline comment; it must be promoted to the body.

360? - `description` -- from the `Description:` field

- `suggestion` -- from the `Suggestion:` field (may be empty)

**\*\*Early-exit on zero issues\*\***: if the parsed issue list is empty, do NOT walk the diff, do NOT build a payload, and do NOT post anything to GitHub. Posting an empty PENDING review is wasteful, looks spammy on the PR, and gives the user a "submit your review" reminder for a review with nothing in it. Instead:

- Print: "Reviewer found no issues on PR #\${pr\_number}. No PENDING review created."
- Skip directly to Step 4 (Cleanup) and then Step 5 (Final report). The Final report should note that no PENDING review was posted.

2. Read `` via `read\_file` and walk it to determine which `(file, line)` pairs are present on the RIGHT side of any hunk. The line is on the RIGHT side when:

- It is an added line (prefix `+`, but not the `+++ b/...` header line), OR
- It is a context line (prefix ` `).

370?

Walk the diff file by file:

- Each file section starts with `+++ b/<path>` (treat `+++ /dev/null` as a deletion -- skip).
- Within a file, each hunk starts with `@@ -<old>[,<oldcount>] +<new>[,<newcount>] @@`. The `,<count>` parts are optional and default to `1` when omitted (so `@@ -42 +42 @@` is a valid single-line hunk that `git diff` and `gh pr diff` both emit). A regex like `^@@ -(\d+)(?:,\d+)? \+(\d+)(?:,\d+)? @@` matches both forms; capture `` from the second group. Reset the right-

side line counter to ``<new>``. (The regex is intentionally unanchored at the end -- real diffs frequently carry trailing context after the second `@@`, e.g. `@@ -10,5 +15,3 @@ def my\_function():`, and `re.match` succeeds on the prefix without a `\$` anchor. Do NOT add `\$`; it would reject every hunk header that includes a function/class hint.)

- Walk the hunk body. For ` ` (context) and `+` lines (excluding `+++` headers), record the current right-side line and increment the counter. For `-` lines (excluding `---` headers), do not increment the right-side counter.

- Lines starting with ` ` (a literal backslash followed by a space) are diff metadata, not file content -- e.g., ` ` No newline at end of file`. Skip them: do not increment the right-side counter, and do not contribute a `(file, line)` pair.

Build a set of `(file, line)` pairs.

3. Partition the parsed issues into two groups:

380? - **Inline comments**: issues whose `(file, line)` is in the diff set.  
- **Promoted to body**: issues whose `(file, line)` is missing or not in the diff set.

4. Build the JSON payload via the `write` tool, saving to ``<pending_review_payload>``:

```
```json
{
  "commit_id": "<head_sha>",
  "body": "<assembled body, see below>",
  "comments": [
390?   {
      "path": "src/foo.rs",
      "line": 42,
      "side": "RIGHT",
      "body": "***[bug] description text\n\n**Suggestion:** suggestion text"
    }
  ]
}
```
```

400? **Do NOT include the `event` field.** Omitting `event` causes the GitHub API to create the review in PENDING state, which is exactly what we want -- the user reviews and submits it through the GitHub UI.

The top-level `body` is constructed as:

```
```
## Summary

<verbatim text of the ## Summary section from the review_file>

## Issue counts by severity
410? - bugs: <X>
    - suggestions: <Y>
    - nits: <Z>

<if any promoted issues exist, append:>
## Issues outside the diff
```
```

These findings reference lines that are not present in the diff and could not be posted as inline comments:

```
420? - [severity] file:line -- description
 - Suggestion: suggestion text
 <repeat for each promoted issue>
 <end if>
```
```

Each inline comment body has the form `[severity] description\n\nSuggestion: suggestion`. If the suggestion is empty, drop the suggestion line.

****Construct the payload as a Python dict, then serialize via `json.dumps`.**** Do NOT concatenate JSON strings by hand: review text routinely contains `"`, `\\`, or embedded newlines, and naive concatenation produces invalid JSON which `gh api` rejects with 400/422. Run a short `python3` heredoc to materialize the JSON string and persist it to `` through the `write` tool:

```
430?  ```bash
python3 <<'PY'
import json
payload = {
    "commit_id": "<head_sha>",
    "body": "<assembled body>",
    "comments": [
        {
            "path": "src/foo.rs",
            "line": 42,
440?         "side": "RIGHT",
            "body": "***[bug]** description text\n\n**Suggestion:** suggestion text",
        },
        # ... one entry per inline comment
    ],
}
print(json.dumps(payload, ensure_ascii=False, indent=2))
PY
```
```

450? `json.dumps(payload, ensure\_ascii=False, indent=2)` handles all quote, backslash, and newline escaping automatically. Capture the printed JSON and pass it as the `content` argument to the `write` tool with `` as the file path.

5. Post the review:

```
```bash
gh api "repos/{owner}/{repo}/pulls/{pr_number}/reviews" \
-X POST \
--input "${pending_review_payload}" \
> /tmp/grok-review-post-${REVIEW_ID}.json 2> /tmp/grok-review-post-${REVIEW_ID}.err
```
```

460? Capture both stdout and stderr.

6. **\*\*Error handling\*\***: if the `gh api` call exits non-zero, surface the captured stderr verbatim to the user along with the HTTP status code (parseable from `gh api`'s stderr message). Do not retry. Common cases by status:

- **\*\*422 Unprocessable Entity\*\***: comments reference lines outside the diff (the diff-line filter missed something, e.g., the reviewer hallucinated a file path), `commit\_id` is stale (the PR was force-pushed between Step 1 and now), or the `side` value was rejected.
- **\*\*403 Forbidden\*\***: authenticated user lacks permission to comment on the PR (read-only collaborator on the repo, archived repo).
- **\*\*404 Not Found\*\***: PR does not exist (the user passed a wrong number, or the URL is in a private repo the user cannot see).
- **\*\*5xx\*\***: GitHub-side outage or transient failure.

On any error, also keep `` on disk (skip the PR-mode review\_file deletion in Step 4) so the user can see what would have been posted and either re-run with corrected inputs or submit the notes through some other channel. Mention the preserved path in the error message.

470? 7. On success, the user-facing pending-review URL is the PR Files tab, where pending reviews surface and where the "Finish your review" / "Submit review" button lives:

```
```
https://github.com/<owner>/<repo>/pull/<pr_number>/files
```
```

Construct this URL from the already-validated `owner`, `repo`, and `pr\_number` -- do NOT use

the `html\_url` field from the GitHub response. The response's `html\_url` points at a deep link to the review object that does not surface the submit button as cleanly as the Files tab. (The response is still parsed for the review `id`, which is recorded in the Final report; the `html\_url` field is not used.)

Print to the user as a structured block (not a wall of text):

480?

...

```
A PENDING review has been created on PR #<pr_number>.
- Inline comments: <N>
- Body findings (outside the diff): <M>
- Submit at: https://github.com/<owner>/<repo>/pull/<pr_number>/files
 (scroll to "Finish your review" -> click "Submit review").
Until you submit, the comments are visible only to you.
...
```

490?## Step 4: Cleanup

The cleanup behavior is **asymmetric** by mode and by outcome:

- **Local and branch modes**: keep ``<review_file>`` and ``<summary_file>`` -- they are the deliverable. Remove only ``<diff_file>`` and the ``/tmp/grok-review-files-${REVIEW_ID}.txt`` helper file.
- **PR mode, successful post**: remove ``<review_file>``, ``<diff_file>``, ``<pending_review_payload>``, ``/tmp/grok-review-prmeta-${REVIEW_ID}.json``, ``/tmp/grok-review-files-${REVIEW_ID}.txt``, and the post stdout/stderr capture files. The pending review on GitHub is the deliverable; the local files are no longer needed. ``<summary_file>`` was never written in PR mode, so no action there.
- **PR mode, failed post (any non-zero `gh api` exit)**: keep ``<review_file>`` so the user can see what would have been posted and submit it through some other channel. Remove the rest as in the success path. The Final report must mention the preserved path.
- **PR mode, no reviewer output (covers both zero-issue early-exit AND PR-with-empty-diff exit)**: remove all of ``<diff_file>``, ``<pending_review_payload>`` (never created -- `rm -f` is a no-op), ``/tmp/grok-review-prmeta-${REVIEW_ID}.json``, ``/tmp/grok-review-files-${REVIEW_ID}.txt``, and ``<review_file>`` (which is a no-op in the empty-diff case where the reviewer never ran, and removes the empty-of-actionable-content file in the zero-issue case). This sub-case is the catch-all for any PR-mode exit that did not POST to GitHub.

Cleanup commands:

500?

```bash

Local / branch mode (always -- covers both successful-review and empty-diff-exit cases; the empty-diff exit happens before `<diff_file>` contains anything useful, but `rm -f` is a no-op on a non-existent or already-empty file)

```
rm -f "${diff_file}" /tmp/grok-review-files-${REVIEW_ID}.txt
```

PR mode, successful post

```
rm -f "${review_file}" "${diff_file}" "${pending_review_payload}" \
    /tmp/grok-review-prmeta-${REVIEW_ID}.json \
510?    /tmp/grok-review-files-${REVIEW_ID}.txt \
    /tmp/grok-review-post-${REVIEW_ID}.json \
    /tmp/grok-review-post-${REVIEW_ID}.err
```

PR mode, failed post -- omit `"${review_file}"` from the rm command

```
rm -f "${diff_file}" "${pending_review_payload}" \
    /tmp/grok-review-prmeta-${REVIEW_ID}.json \
    /tmp/grok-review-files-${REVIEW_ID}.txt \
    /tmp/grok-review-post-${REVIEW_ID}.json \
    /tmp/grok-review-post-${REVIEW_ID}.err
```

520?

PR mode, no reviewer output (zero-issue early-exit OR empty-diff exit)


```
rm -f "${review_file}" "${diff_file}" "${pending_review_payload}" \
    /tmp/grok-review-prmeta-${REVIEW_ID}.json \
    /tmp/grok-review-files-${REVIEW_ID}.txt
...
```

Step 5: Final report

Present a final report to the user. Item 1 is always included; item 2 is omitted on empty-diff exits; item 3 is omitted on empty-diff exits and on PR zero-issue early-exits; item 4 is always included.

530?

1. ****Mode + target**** -- e.g., "Local changes" or "Branch feature/foo vs origin/main" or "PR #4221: <pr_title>".
 2. ****Files reviewed**** -- count and (if 10 or fewer) the list. Omit on empty-diff exits.
 3. ****Issues by severity**** -- the bug/suggestion/nit counts. Omit on empty-diff exits and on PR zero-issue early-exits (counts are all zero by construction; the mode-specific bullet says so).
 4. ****Mode-specific output****:
 - Local / branch (successful review): full paths to <review_file> and <summary_file>, plus a one-line copy of the summary's top-issues list.
 - Local / branch (empty-diff exit): "No changes to review." (local mode) or "Branch <target> has no changes vs <BASE>." (branch mode). No file paths -- nothing was written.
 - PR (successful post): the structured submit block from Step 3 PR mode item 7 (PR number, inline-comment count, body-findings count, Files-tab URL, submit reminder), plus the review `id` from the response (for traceability).
 - PR (zero-issue early-exit): "Reviewer found no issues on PR #<pr_number>. No PENDING review created."
 - PR (empty-diff exit): "PR #<pr_number> has no changes. Nothing to review."
- 540? - PR (failed post): the verbatim stderr from `gh api` plus the HTTP status code, plus the preserved path to <review_file> so the user can recover the findings.

In-Progress Reporting

Give the user a brief status update after each phase:

- After argument parsing: "Reviewing <mode> target: <target-summary>." (omit target for local mode -- "Reviewing local changes.")
- After Step 1 (diff collection): "Collected diff (<N> files, <M> changed lines). Launching reviewer..."
- After Step 1 (empty diff): "No changes to review. Proceeding to final report."
- After Step 2 (reviewer complete): "Review complete. Processing findings..."
- 550?- After Step 3 (local / branch): "Found N issues (X bugs, Y suggestions, Z nits). Wrote <review_file> and <summary_file>."
- After Step 3 (PR, zero issues): "Reviewer found no issues on PR #<pr_number>. No PENDING review created."
- After Step 3 (PR, success): "Posted PENDING review with N inline comments and M body findings. Visit <url> to submit."
- After Step 3 (PR, error -- 422 or any other non-zero exit): "Failed to post pending review. HTTP <status>. GitHub returned: <verbatim stderr>. Review notes preserved at <review_file>."

Rules

- ****Reviewer is read-only**** -- the reviewer subagent must never modify files. The orchestrator must never modify source files either. The only writes are to <review_file>, <summary_file>, <diff_file>, <pending_review_payload>, and the GitHub PENDING review.
- ****Inject the reviewer persona into the prompt**** -- always prepend the `reviewer` persona instructions (from the shared personas file) to the subagent prompt. Do NOT pass a `persona` parameter to `spawn\_subagent`.
- ****Prefix `description` with `[reviewer]\*\*`** -- the pager's subagent label renderer parses the first `[tag]` of the description and uses it as the row label (the bracketed prefix is stripped from the displayed description). Without it, the row falls through to the generic "General" label because `subagent\_type` is `general-purpose` and no persona/role is plumbed through the spawn args.
- 560?- ****Include context in the reviewer prompt**** -- the reviewer needs the conversation context (the user's framing, any constraints, etc.) passed through the task prompt.

- **One reviewer per run** -- this skill runs a single reviewer. Multi-reviewer runs are the job of `/implement` with `--effort N`.
- **Disambiguation is deterministic** -- the auto-detect rules in argument parsing are applied in order. Do not second-guess them. If the rules do not match, ask the user; do not guess.
- **Empty diffs short-circuit** -- never launch the reviewer with an empty diff. Report "no changes", run cleanup, and produce a Final report (using the appropriate "empty-diff exit" bullet).
- **Zero issues short-circuits PR mode** -- if the reviewer finds nothing on a PR, do not post an empty PENDING review. Print a "no issues found" message, clean up, and stop.
- **PR mode requires `gh` auth** -- if `gh auth status` fails, stop with a clear instruction to run `gh auth login`. Do not try to work around it.
- **PENDING reviews are the user's to submit** -- the orchestrator never sets the `event` field on the review payload. The user reviews and submits via the GitHub UI. Always print the URL plus the submit-via-UI reminder.
- **Filter inline comments to lines actually in the diff** -- GitHub returns 422 for comments on lines outside the diff. The orchestrator parses the diff itself to validate `(file, line)` pairs and promotes any out-of-diff issues to the top-level body as bullet points.
- **Surface every `gh api` error verbatim** -- if the GitHub API call exits non-zero (422, 403, 404, 5xx, anything else), do not retry. Show the user the raw stderr plus the HTTP status code so they can debug, and preserve `<review_file>` on disk so the findings are not lost.
- **No helper script** -- the orchestrator does the diff parsing and JSON building inline.
- **Cleanup is asymmetric by mode and outcome** -- local/branch modes keep `<review_file>` and `<summary_file>` (they are the deliverable). PR mode removes `<review_file>` on a successful post (the pending GitHub review is the deliverable) and on a no-reviewer-output exit (zero issues OR empty diff -- nothing actionable to preserve), but preserves it on a failed post so the user can recover the findings.
- **Thread the same file paths** across all steps -- never regenerate paths between steps. The `REVIEW_ID` is fixed for the run.
- **Error handling** -- if the reviewer subagent fails, the diff collection fails, or the GitHub API call fails, report the error to the user and stop. Do not silently continue with missing results.
- **No emojis in output** -- match the conventions of the `reviewer` persona instructions and the surrounding skills.

Design notes

These are background notes that explain the rationale behind some of the choices above. They are NOT rules; they are pointers for future maintainers.

- **No helper script.** The markdown parsing of `<review_file>` and the unified-diff hunk walk are well-defined and small enough to do directly in the orchestrator (using `read_file` to load the inputs and `write` to create the JSON payload). This follows the same self-contained pattern as `xai-pr-comments`. If a future maintainer finds the inline approach genuinely intractable -- e.g., the parser is consistently miscounting hunks across many real-world PRs -- a helper at `.grok/skills/review/scripts/build_pending_review.py` could be added with a clear justification. As of this writing, no such helper exists.
- **Owner/repo are derived from the PR url**, not from a `baseRepository` field. `gh pr view --json baseRepository` returns `Unknown JSON field: "baseRepository"` -- only `headRepository`, `headRepositoryOwner`, and `isCrossRepository` are exposed. Parsing the URL is the simplest universal approach and works correctly for cross-repo PRs.
- **The constructed `/files` URL is used in user output, not the response's `html_url`.** The Files tab is where the "Finish your review" / "Submit review" button surfaces in the GitHub UI; that is what the user actually needs to interact with. The response's `html_url` deep-links to the review object and is less useful for this workflow.

9. tool_result

```
exit: 0
{"additions":1487,"author":{"id":"MDQ6VXNlcjE2NzQ4Nzc=","is_bot":false,"login":"avidullu","name":"Avi"},"baseRefName":"master","body":"## Summary\n\nTwo things, both about **release confidence**:\n\n1. **New ? `scripts/gates/perf/`:** a standalone, release-gating performance tool that compares the binary **deployed on `api.khelsutra.guru`** against **repo HEAD** , on the local GPU, with no SPA / no user step. It times the four release-critical stages independently ? **checksum ? validation ? segmentation (`wasb_hybrid`, GPU via WSL) ? full-reel stitch** ? and
```

gates the release (exit 1 on a stitch/total regression beyond tolerance).\n2. **Reorg ?**
consolidate the scattered nightly CLIs into `scripts/gates/`: **a cohesive gate package** (quality + performance legs) instead of loose files, honoring the `docs/CODE\_MAP.md` \"standalone CLIs live in `scripts/`\" convention.\n\n## The perf tool\n- **Prod resolution:** live Cloud Run image digest ? `perf\_manifest.json` digest?commit map ? pinned ? `--prod-commit`. (Confirmed live: `sha256:7b62a8df? ? 304af7b`.)\n- **Fair comparison:** each commit runs from its own throwaway `git worktree`, one canonical config, identical inputs ? only the `code` differs. `skip\_ai\_handoff=true` + fast `stream\_video` decode = deterministic + cloud-free. Always stitches the entire reel (watermark on, libx264 both sides).\n- **Self-help preflight** names each missing prereq + its fix. Flags: `--preflight`, `--stitch-only`, `--profile quick|full|stress`, `--update-baseline`.\n- **Exit codes** match the quality gates: `0 PASS` . 1 REGRESSION . 2 nothing-ran/preflight . 4 STALE.\n- Videos referenced by `basename + --vault-dir` (default `~/Videos`) so no personal path is committed (leak-scan guard).\n\n### Nightly integration\n`scripts/gates/nightly\_perf.ps1` mirrors `nightly\_regression.ps1`;
`register\_nightly\_task.ps1` now installs **both** `KhelSutraNightlyRegression` (quality, 30-min cap) and `KhelSutraNightlyPerf` (perf, 3-hr cap ? a separate task since a GPU run far exceeds the quality cap).\n\n## The reorg\n\n`scripts/nightly\_regression.py` / `nightly\_reelcount.py` / `nightly\_email.py` ?
`scripts/gates/quality/scripts/nightly\_regression.ps1` / `register\_nightly\_task.ps1`
? `scripts/gates/` + `scripts/gates/perf/` (new) + `scripts/gates/{README.md, nightly\_perf.ps1}`\n\nMovers preserve git history. **Every** reference updated in-tree: the deployed GCP nightly (`run\_on\_vm.sh` import + invocation), `tests/test\_nightly\_` loaders, `eval\_baselines/` docstrings, and the active docs (`CODE\_MAP`, `NIGHTLY\_ENGINE\_REGRESSION`, `QUALITY\_ITERATIONS`, `docs/README`, `ALPHA\_LAUNCH\_READINESS`, `DECISION\_SNAPSHOT\_REGRESSION`). Historical `docs/archives/` left as-is.\n\n## What the tool found (this run ? RTX 2070, prod `304af7b` vs HEAD `3249451`, 3 clips)\n\n**Performance-neutral, no regression.** checksum/validation/segmentation unchanged (same detector/inputs). The one code-level delta ? `#531` single-pass stitch ? lands **within run-to-run noise**: matrix stitch ? was ?10% / ?13% / +5% across clips, and the controlled A/B (median of reps) was 1080p **+6.5%** vs 5.3K **+9.8%** (directions disagree). `#531`'s real win is the L4 **NVENC** path, not reproducible with local libx264. The other 8 release commits are cloud-only.\n\n## Verification\n- `ruff check` (pinned `0.15.16`) ? . `mypy backend training` ? . full suite **1933 passed / coverage 84.28%** (floor 70) ?\n- Tool **self-tested end-to-end** (preflight 9/9, prod auto-resolve, worktrees, stitch A/B, gating, report, exit 0). The self-test caught + fixed two Windows bugs in the tool: gcloud `CMD` resolution, and worktree `MAX\_PATH` (added `core.longpaths=true` + a temp worktree root).\n- ?? 2 pre-existing `test\_storage.py` failures (GCSFuse `#531`/`#535`) are **Windows path-separator** issues unrelated to this change ? they pass on CI Linux (master is green). Filed separately.\n\n## Caveats\n- Local, in-process measurement ? cloud-topology w ins in the release (GCSFuse mount `#535`, CP `acquire\_local\_input` gating `#539`, cancel `#538`) can't be exercised locally.\n- 8 GB RTX 2070 vs prod's 24 GB L4: absolute segmentation times aren't comparable to prod; the prod-vs-HEAD **delta on one box** is the signal.\n- `--update-baseline` seeds `eval\_baselines/perf\_baseline.json` on the owner box (machine-specific, like the quality baselines ? not committed).\n\n? Generated with [Claude Code](https://claude.com/claude-code)\n", "commits": [{"authoredDate": "2026-07-09T04:20:27Z", "authors": [{"email": "avi.dullu@gmail.com", "id": "MDQ6VXNlcjE2NzQ4Nzc=", "login": "avi dullu", "name": "Avi Dullu"}, {"email": "noreply@anthropic.com", "id": "MDQ6VXNlcjgxDQ3", "login": "claude", "name": "Claude Opus 4.8"}], "committedDate": "2026-07-09T04:20:27Z", "messageBody": "?ts/gates/\n\nAdd scripts/gates/perf/ ? a standalone, release-gating tool that times\nchecksum -> validation -> segmentation -> full-reel-stitch for the commit\nndeployed on api.khelsutra.guru vs repo HEAD, on the local GPU (wasb_hybrid via\nWSL), and gates the release (exit 1 on a stitch/total regression beyond\ntolerance). Preflight self-help, prod auto-resolve from the live Cloud Run\nimage, isolated git worktrees, dated reports under output/nightly_perf/.\n\nConsolidate the scattered nightly CLIs into a cohesive scripts/gates/ package\n(honoring the docs/CODE_MAP.md \"standalone CLIs live in scripts/\" convention):\n\n nightly_regression.py / nightly_reelcount.py / nightly_email.py -> gates/quality/\n nightly_regression.ps1 -> gates/ ; register_nightly_task.ps1 -> gates/ (now\n registers BOTH the quality and the new performance scheduled tasks).\n\nAll references updated in-tree: the deployed GCP nightly (run_on_vm.sh import +\ninvocation), tests, eval-baseline docstrings, and the active docs (CODE_MAP,\nNIGHTLY_ENGINE_REGRESSION, QUALITY_ITERATIONS, README, ...). Archives left as-is.\n\nVerified: ruff 0.15.16 clean, mypy clean, full suite 1933 passed / coverage\n84.28%. (2 pre-existing Windows-only test_storage GCSFuse #535 failures are\nunrelated to this change and pass on CI Linux.) Tool self-tested end-to-end.\n\nCo-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>\", \"messageHeadline\": \"feat(gates): local release perf gate + consolidate

nightly into scrip?", "oid": "0aa6d4b06a2cd82be07c25205df30109e8fca61e"}], "deletions": 113, "files": [{"path": "deploy/gcp/nightly_regression/README.md", "additions": 8, "deletions": 8}, {"path": "deploy/gcp/nightly_regression/run_on_vm.sh", "additions": 3, "deletions": 3}, {"path": "docs/ALPHA_LAUNCH_READINESS.md", "additions": 4, "deletions": 4}, {"path": "docs/CODE_MAP.md", "additions": 5, "deletions": 5}, {"path": "docs/DECISION_SNAPSHOT_REGRESSION.md", "additions": 4, "deletions": 4}, {"path": "docs/NIGHTLY_ENGINE_REGRESSION.md", "additions": 6, "deletions": 6}, {"path": "docs/QUALITY_ITERATIONS.md", "additions": 2, "deletions": 2}, {"path": "docs/README.md", "additions": 1, "deletions": 1}, {"path": "eval_baselines/nightly_baseline.json", "additions": 1, "deletions": 1}, {"path": "eval_baselines/perf_manifest.json", "additions": 40, "deletions": 0}, {"path": "eval_baselines/reelcount_manifest.json", "additions": 1, "deletions": 1}, {"path": "scripts/gates/README.md", "additions": 75, "deletions": 0}, {"path": "scripts/gates/nightly_perf.ps1", "additions": 80, "deletions": 0}, {"path": "scripts/gates/nightly_regression.ps1", "additions": 2, "deletions": 2}, {"path": "scripts/gates/perf/README.md", "additions": 139, "deletions": 0}, {"path": "scripts/gates/perf/config/bench_config.json", "additions": 64, "deletions": 0}, {"path": "scripts/gates/perf/harness.py", "additions": 335, "deletions": 0}, {"path": "scripts/gates/perf/preflight.py", "additions": 120, "deletions": 0}, {"path": "scripts/gates/perf/run.py", "additions": 397, "deletions": 0}, {"path": "scripts/gates/perf/stitch_ab.py", "additions": 94, "deletions": 0}, {"path": "scripts/gates/quality/nightly_email.py", "additions": 0, "deletions": 0}, {"path": "scripts/gates/quality/nightly_reelcount.py", "additions": 2, "deletions": 2}, {"path": "scripts/gates/quality/nightly_regression.py", "additions": 1, "deletions": 1}, {"path": "scripts/gates/register_nightly_task.ps1", "additions": 98, "deletions": 0}, {"path": "scripts/register_nightly_task.ps1", "additions": 0, "deletions": 70}, {"path": "tests/test_nightly_reelcount.py", "additions": 2, "deletions": 2}, {"path": "tests/test_nightly_regression.py", "additions": 3, "deletions": 1}], "headRefName": "feat/perf-release-gate", "reviewDecision": "", "state": "OPEN", "statusCheckRollup": [{"__typename": "CheckRun", "completedAt": "2026-07-09T04:21:55Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223708", "name": "Ruff lint", "startedAt": "2026-07-09T04:21:47Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:21:56Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223664", "name": "Leak scan (PII / attribution guard)", "startedAt": "2026-07-09T04:21:47Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:21:57Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223673", "name": "Markdown link check (live docs)", "startedAt": "2026-07-09T04:21:47Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:22:24Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223689", "name": "Mypy (lenient baseline)", "startedAt": "2026-07-09T04:21:47Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:22:11Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223650", "name": "Syntax sweep + import smoke (no GPU stack)", "startedAt": "2026-07-09T04:21:47Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:25:02Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223667", "name": "Full test suite (offline, mocked)", "startedAt": "2026-07-09T04:21:47Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:22:13Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223655", "name": "Golden fixtures cross-OS (ubuntu-latest)", "startedAt": "2026-07-09T04:21:47Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:23:03Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223688", "name": "Golden fixtures cross-OS (windows-latest)", "startedAt": "2026-07-09T04:21:50Z", "status": "COMPLETED", "workflowName": "CI"}, {"__typename": "CheckRun", "completedAt": "2026-07-09T04:22:06Z", "conclusion": "SUCCESS", "detailsUrl": "https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223675", "name": "Golden fixtures cross-OS"}]

```
(macos-latest)","startedAt":"2026-07-09T04:21:48Z","status":"COMPLETED","workflowName":"CI"},{"__typename":"CheckRun","completedAt":"2026-07-09T04:22:18Z","conclusion":"SUCCESS","detailsUrl":"https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223663","name":"Dependency license scan (no copyleft in runtime deps)","startedAt":"2026-07-09T04:21:47Z","status":"COMPLETED","workflowName":"CI"},{"__typename":"CheckRun","completedAt":"2026-07-09T04:22:56Z","conclusion":"SUCCESS","detailsUrl":"https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28993843155/job/86039223648","name":"Built wheel smoke (install + launch)","startedAt":"2026-07-09T04:21:47Z","status":"COMPLETED","workflowName":"CI"}],{"title":"feat(gates): local release perf gate + consolidate nightly into scripts/gates/","url":"https://github.com/Khelsutra/badminton-highlight-indexer/pull/547"}
```

10. assistant

Reviewing PR #547: local release perf gate + nightly consolidation into `scripts/gates/`. Setting up the review workflow.

11. tool_result

Error: /home/avidullu/.grok/bundled/shared/personas/reviewer.md does not exist.
Note: your current working directory is /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer

12. tool_result

```
exit: 0
36f03232
github.com
? Logged in to github.com account avidullu (/home/avidullu/.config/gh/hosts.yml)
- Active account: true
- Git operations protocol: https
- Token: gho_*****
- Token scopes: 'gist', 'read:org', 'repo', 'workflow'
```

13. tool_result

```
exit: 0
/home/avidullu/.grok/bundled/skills/shared/personas/reviewer.md
agents
manifest.json
personas
roles
skills
CHANGELOG.json
CHANGELOG.md
README.md
active_sessions.json
active_sessions.lock
agent_id
auth.json
auth.json.lock
bin
bundled
completions
config.toml
docs
downloads
installed-plugins
logs
managed_config.lock
```

marketplace-cache
models_cache.json
projects
sessions
skills
slash-mru.json
tip_cursor.json
upload_queue
vendor
version.json
worktrees.db

14. tool_result

1?You are a meticulous code reviewer. Review code and produce structured review notes in a Markdown file at the path given in the prompt.

Process:

1. Read all relevant code thoroughly
2. Write findings to the specified review notes file
3. Use structured format: severity, file:line, description, suggestion, status

Rules:

- 10?- Check correctness first, style second
- Look for edge cases, error handling gaps, race conditions
 - Flag unwrap(), unnecessary clone(), or lock usage
 - Be specific: cite file:line for every issue
 - Do NOT fix the code yourself
 - In your final response, state the file path and summarize the verdict

15. tool_result

exit: 0

Unknown JSON field: "baseRef0id"

Available fields:

additions
assignees
author
autoMergeRequest
baseRefName
body
changedFiles
closed
closedAt
comments
commits
createdAt
deletions
files
headRefName
headRef0id
headRepository
headRepositoryOwner
id
isCrossRepository
isDraft
labels
latestReviews
maintainerCanModify
mergeCommit
mergeStateStatus
mergeable
mergedAt
mergedBy
milestone

- scripts/gates/quality/nightly_email.py
- scripts/gates/quality/nightly_reelcount.py
- scripts/gates/quality/nightly_regression.py
- scripts/gates/register_nightly_task.ps1
- scripts/register_nightly_task.ps1
- tests/test_nightly_reelcount.py
- tests/test_nightly_regression.py

diff_bytes: 123727

123727 /tmp/grok-review-diff-36f03232.diff

18. assistant

Collected diff (27 files, ~124 KB). Launching the reviewer subagent.

19. tool_result

Subagent started in background.

subagent_id: 019f451f-c9d0-76c2-872f-8da5394c2a2a

type: general-purpose

description: [reviewer] pr #547

Use get_command_or_subagent_output with task_ids=["019f451f-c9d0-76c2-872f-8da5394c2a2a"] and timeout_ms to wait for results.

20. tool_result

=== Task 019f451f-c9d0-76c2-872f-8da5394c2a2a ===

Command: [subagent:general-purpose] [reviewer] pr #547

Status: completed

Started: 2026-07-09T04:25:53Z

Ended: 2026-07-09T04:28:38Z

Duration: 165.11s

Exit Code: 0

Output File:

=== Output ===

Review complete

Structured notes: **~/tmp/grok-review-36f03232.md**

Verdict

Solid reorg + a useful prod-vs-HEAD perf harness (worktrees, shared bench config, `skip\_ai\_handoff`, stitch A/B). Quality moves (REPO_ROOT depth, GCP `run\_on\_vm.sh`, tests) look consistent. **Not merge-clean for defaults/gating** without fixes.

Highest-impact findings

1. **Bug ? `\$RepoRoot` default is one level short** in `register\_nightly\_task.ps1` / `nightly\_perf.ps1` (and the moved quality wrapper): default points at `scripts/`, so registration without `-RepoRoot` never arms.
2. **Bug ? HEAD failures don't fail the gate**: missing stage times ? `pct` is `None` ? exit **0**.
3. **Bug ? empty/stitch-only with no sources still PASSes** (exit 0).
4. **Suggestions**: gate `total\_s` includes noisy segmentation; no pure tests for the new gate core; stale paths in `perf/README.md`; deleting old scheduled-task scripts without shims silently disables existing tasks.

CI green is expected (GPU/Windows paths aren't exercised there).

<subagent_meta>id=019f451f-c9d0-76c2-872f-8da5394c2a2a, type=general-purpose, tool_calls=40, turns=1, duration_ms=165110</subagent_meta>

<subagent_result>

subagent_id: 019f451f-c9d0-76c2-872f-8da5394c2a2a

subagent_type: general-purpose

To continue this subagent's conversation, use
resume_from="019f451f-c9d0-76c2-872f-8da5394c2a2a".
</subagent_result>

21. assistant

Review complete. Processing findings and preparing the PENDING GitHub review.

22. tool_result

1?## Summary

PR #547 consolidates the nightly quality CLIs under `scripts/gates/quality/`, wires GCP `run\_on\_vm.sh` + tests to the new paths, and adds a solid local release **performance** harness (`scripts/gates/perf/`) that compares a resolved prod commit vs HEAD via git worktrees, shared `bench\_config.json`, `skip\_ai\_handoff`, and stage timing. The reorg of quality scripts (REPO_ROOT depth, import path, test loaders, GCP runner) looks coherent. Dominant risks: **broken default `\$RepoRoot` in the new PowerShell wrappers/register script** (one `Split-Path` short after the move), **the perf gate treating hard failures as non-regressions** (missing stages ? `pct` returns `None` ? exit 0), and **silent PASS when nothing measurable ran** (empty stitch/vault). Also: no pure-unit tests for the new gate logic, stale paths in `perf/README.md`, and existing Windows scheduled tasks will no-op after delete-without-shim until re-registered.

Issues

Issue 1 -- Severity: bug

- File: scripts/gates/register_nightly_task.ps1:31
- Description: Default `\$RepoRoot` only walks up two directories (`Split-Path` twice). The script now lives at `scripts/gates/register\_nightly\_task.ps1`, so the default resolves to `?/scripts` rather than the repo root. Running without an explicit `-RepoRoot` registers both tasks with wrong paths (`scripts\scripts\gates\?`), so wrappers permanently hit ?wrapper absent / SKIPPED? and never arm. The same two-level default is copied into `nightly\_perf.ps1` and was left unchanged on the moved `nightly\_regression.ps1`.
- 10?- Suggestion: Use three `Split-Path -Parent` hops (or resolve via `Join-Path \$PSScriptRoot '..\..\.'` and `Resolve-Path`) so default `\$RepoRoot` is the checkout root. Fix the same default in `scripts/gates/nightly\_perf.ps1` and `scripts/gates/nightly\_regression.ps1`.
- Status: open

Issue 2 -- Severity: bug

- File: scripts/gates/perf/run.py:165
- Description: Gating only fires when both sides produce numeric stage times (`d` is not None). If HEAD fails (no segments ? `stitch\_s=None`, harness crash ? empty result, timeout ? missing JSON), `pct()` returns `None` and **no regression finding is recorded**. A completely broken HEAD can still exit 0 PASS while the report quietly lists errors. Quality gates treat pipeline errors as regressions; this gate does not.
- Suggestion: In `build()` / `exit\_code()`, treat missing/failed gated stages on `rc` (when `prod` succeeded) as regression (or at least exit 2). Also fail if `row["errors"]["rc"]` is non-empty for a gated path. Mirror quality?s ?errored video = regression? semantics.
- Status: open

Issue 3 -- Severity: bug

20?- File: scripts/gates/perf/run.py:313

- Description: The only ?nothing to run? guard is `if not videos and not args.stitch\_only`. With `--stitch-only` (or when the matrix is skipped and every `stitch\_ab` source is missing), the driver still builds an empty summary, produces no findings, and returns **0 PASS**. A misconfigured vault / profile therefore green-lights the release.
- Suggestion: After the run loop, if there are no comparable stage rows and no completed stitch A/B medians, return exit 2 (`nothing-ran`). Do the same when every harness/stitch result is errors-only.
- Status: open

Issue 4 -- Severity: bug

- File: scripts/gates/nightly_perf.ps1:29

- Description: Same incorrect default ``$RepoRoot`` as Issue 1 (``Split-Path`` ×2 from ``scripts/gates/``). Manual invocation without ``-RepoRoot`` looks for ``scripts\scripts\gates\perf\run.py``, takes the `?tool absent ? exit 0 SKIPPED?` branch, and never runs the gate.
- Suggestion: Default ``$RepoRoot`` to three parents of ``$PSScriptRoot`` / ``$MyInvocation.MyCommand.Path``. Prefer requiring callers to pass ``-RepoRoot`` only when non-default.
- Status: open

30?

Issue 5 -- Severity: suggestion

- File: `scripts/gates/perf/run.py:190`
- Description: ``exit_code()`` never returns 3, and a missing ``perf_baseline.json`` is not treated as `?no baseline?` Shared suite docs (``scripts/gates/README.md``, wrapper status ``NOBASELINE``) claim the common exit family including `**3 = no baseline**`, and ``nightly_perf.ps1`` maps code 3 to ``NOBASELINE``. Live prod-vs-HEAD comparison can work without a frozen baseline, but the documented contract is inconsistent and the wrapper's branch is dead.
- Suggestion: Either implement exit 3 when baseline gating is expected and the file is missing, or document that perf only uses 0/1/2/4, remove the wrapper's ``3`` branch, and drop `?3 no baseline?` from the shared exit-code table.
- Status: open

Issue 6 -- Severity: suggestion

- File: `eval_baselines/perf_manifest.json:20`
- Description: Gate stages include both ``stitch_s`` and ``total_s``. ``total_s`` is the sum of checksum + validation + `**segmentation**` + stitch. Segmentation is explicitly described as a constant backdrop / not gated, but it usually dominates wall time and is GPU/WSL-noisy. A 20% swing in segmentation alone can fail the release on ``total_s`` even when stitch (the intended #531 signal) is flat or improved?undermining the fairness story.
- 40?- Suggestion: Gate only ``stitch_s`` (+ always-on stitch A/B). Keep ``total_s`` as report-only context, or require both stitch regression *and* total regression before failing. Document why total is gated if you keep it.
- Status: open

Issue 7 -- Severity: suggestion

- File: `scripts/gates/perf/run.py:332`
- Description: No unit tests cover the pure gate core (``pct``, ``build``, ``exit_code``, stale detection, prod resolve fallback). Quality nightlies ship offline pure-core tests; this PR only rewires those paths and adds a large untested harness. Easy-to-regress bugs (Issues 2?3) would have been caught by a few dict-in/dict-out tests.
- Suggestion: Add ``tests/test_perf_gate.py`` loading ``run.py`` by path (same pattern as ``test_nightly_regression.py``) and covering: regression/improvement thresholds, missing-stage handling, empty results `? exit 2`, stale prod_commit/profile, ``pct`` edge cases (``prod==0``, ``None``).
- Status: open

Issue 8 -- Severity: suggestion

- 50?- File: `scripts/gates/perf/README.md:96`
- Description: Nightly integration / Files sections still document pre-move paths (``scripts/nightly_perf.ps1``, ``scripts/register_nightly_task.ps1``) and quality siblings as ``scripts/nightly_regression.py`` / ``scripts/nightly_reelcount.py``. Operators following this README will run missing scripts. ``scripts/gates/README.md`` has the correct paths; this file was not updated to match the reorg.
- Suggestion: Point every operator path at ``scripts/gates/?`` and align the companion quality references with ``scripts/gates/quality/?``.
- Status: open

Issue 9 -- Severity: suggestion

- File: `scripts/gates/README.md:68`
- Description: Old ``scripts/register_nightly_task.ps1`` and ``scripts/nightly_regression.ps1`` are deleted with no shim. Existing Windows Scheduled Tasks encode the `**old**` wrapper path at registration time; after merge they hit the inline `?wrapper absent ? SKIPPED` exit 0? guard forever and stop protecting quality until someone re-runs the new register script. Easy to miss in production.
- Suggestion: Leave thin shims at the old paths that forward to ``scripts/gates/?``, and call out

in the gates README / PR body: ?re-run `pwsh -File scripts/gates/register_nightly_task.ps1

-RepoRoot ?` after merge.?

- Status: open

60?

Issue 10 -- Severity: suggestion

- File: scripts/gates/nightly_perf.ps1:73

- Description: Exit code 2 (preflight failed / env not ready) is rewritten to `**exit 0 SKIPPED**`. Intentional ?don?t red-light Task Scheduler before WSL lands,? but a lasting env break (GPU driver, missing vault, broken wasb env) also looks green every night. Task history will not surface failures unless someone reads `LATEST\_STATUS.txt`.

- Suggestion: Keep writing `SKIPPED` to the status file, but exit non-zero after N consecutive skips, or use a distinct scheduled-task result / email on repeated preflight failure. At least document that Task Scheduler ?success? is not the gate signal.

- Status: open

Issue 11 -- Severity: suggestion

- File: scripts/gates/perf/run.py:336

- Description: Comparison order is always prod then rc for every video. Shared machine effects (disk cache, GPU clocks, thermal throttle, WSL cold start) systematically favor the second run (HEAD). That biases the ?HEAD slower than prod? gate toward false improvements / missed regressions.

70?- Suggestion: Alternate order per video (or randomize with a logged seed), add a short warmup stitch/segment, and/or require N reps with median like stitch A/B for the full matrix?s gated stages.

- Status: open

Issue 12 -- Severity: nit

- File: scripts/gates/perf/run.py:11

- Description: Module docstring still says companions are `scripts/nightly\_regression.py` + `scripts/nightly\_reelcount.py` (pre-move). Same class of path drift as the perf README.

- Suggestion: Update to `scripts/gates/quality/nightly\_regression.py` and `~/nightly\_reelcount.py`.

- Status: open

Issue 13 -- Severity: nit

80?- File: scripts/gates/perf/harness.py:114

- Description: After a successful `process\_video`, stitch inputs are taken from `db.query\_play\_segments` with a fallback to the in-memory return value. There is no `random.seed(0)` (reelcount uses it for determinism) and no guard that prod/rc segment *counts* match before comparing `stitch\_s`. If HEAD changes segmentation, stitch times are not an apples-to-apples code-path comparison (stitch A/B with fixed pairs is the fair path; the full-matrix stitch gate is not).

- Suggestion: Prefer fixed pairs for any gated stitch timing (or skip gating full-matrix `stitch\_s` when `segments.prod != segments.rc`). Optionally seed RNGs for less noise on secondary fields.

- Status: open

23. tool_result

exit: 0

right_side_pairs: 1709

scripts/gates/perf/run.py: 397 lines

scripts/gates/perf/harness.py: 335 lines

scripts/gates/perf/README.md: 139 lines

scripts/gates/perf/preflight.py: 120 lines

scripts/gates/register_nightly_task.ps1: 98 lines

scripts/gates/perf/stitch_ab.py: 94 lines

scripts/gates/nightly_perf.ps1: 80 lines

scripts/gates/README.md: 75 lines

scripts/gates/perf/config/bench_config.json: 64 lines

deploy/gcp/nightly_regression/README.md: 45 lines

eval_baselines/perf_manifest.json: 40 lines

docs/NIGHTLY_ENGINE_REGRESSION.md: 36 lines

docs/ALPHA_LAUNCH_READINESS.md: 28 lines

docs/DECISION_SNAPSHOT_REGRESSION.md: 25 lines
docs/CODE_MAP.md: 23 lines

24. tool_result

```
exit: 0
OK: scripts/gates/register_nightly_task.ps1:31 [bug]
OK: scripts/gates/perf/run.py:165 [bug]
OK: scripts/gates/perf/run.py:313 [bug]
OK: scripts/gates/nightly_perf.ps1:29 [bug]
OK: scripts/gates/perf/run.py:190 [suggestion]
OK: eval_baselines/perf_manifest.json:20 [suggestion]
OK: scripts/gates/perf/run.py:332 [suggestion]
OK: scripts/gates/perf/README.md:96 [suggestion]
OK: scripts/gates/README.md:68 [suggestion]
OK: scripts/gates/nightly_perf.ps1:73 [suggestion]
OK: scripts/gates/perf/run.py:336 [suggestion]
OK: scripts/gates/perf/run.py:11 [nit]
OK: scripts/gates/perf/harness.py:114 [nit]
```

```
inline=13 promoted=0
right: scripts/gates/nightly_regression.ps1:3
right: scripts/gates/nightly_regression.ps1:4
right: scripts/gates/nightly_regression.ps1:5
right: scripts/gates/nightly_regression.ps1:6
right: scripts/gates/nightly_regression.ps1:7
right: scripts/gates/nightly_regression.ps1:8
right: scripts/gates/nightly_regression.ps1:9
right: scripts/gates/nightly_regression.ps1:45
right: scripts/gates/nightly_regression.ps1:46
right: scripts/gates/nightly_regression.ps1:47
right: scripts/gates/nightly_regression.ps1:48
right: scripts/gates/nightly_regression.ps1:49
right: scripts/gates/nightly_regression.ps1:50
right: scripts/gates/nightly_regression.ps1:51
```

25. assistant

All 13 findings map to valid diff lines. Building and posting the PENDING review.

26. tool_result

```
exit: 0
wrote /tmp/grok-review-pending-36f03232.json (10317 bytes)
comments: 13
bugs=4 suggestions=7 nits=2
exit=0
id= 4659798341
state= PENDING
html_url= https://github.com/Khelsutra/badminton-highlight-
indexer/pull/547#pullrequestreview-4659798341
```

27. tool_result

```
exit: 0
cleaned
```

28. assistant

Review complete ? PR #547

```
**PR:** [feat(gates): local release perf gate + consolidate nightly into
scripts/gates/](https://github.com/Khelsutra/badminton-highlight-indexer/pull/547)
**Scope:** 27 files (local perf release gate + nightly reorg under `scripts/gates/`)
```

CI: all green

Issues by severity

| Severity | Count |
|------------|-------|
| bug | 4 |
| suggestion | 7 |
| nit | 2 |

Highest-impact findings

1. **Bug** ? `$RepoRoot` default is one level short in `register_nightly_task.ps1` / `nightly_perf.ps1` (and the moved quality wrapper). Default resolves to `scripts/` instead of repo root, so registration without `-RepoRoot` never arms correctly.
2. **Bug** ? HEAD failures don't fail the gate: missing stage times ? `pct` is `None` ? exit `0` PASS.
3. **Bug** ? empty / stitch-only with no sources still PASSES (exit 0).
4. **Suggestions:** gate `total_s` includes noisy segmentation; no pure unit tests for the gate core; stale paths in `perf/README.md`; deleting old scheduled-task scripts without shims silently disables existing tasks.

PENDING review posted

A **PENDING** review is ready for you to submit:

- **Inline comments:** 13
- **Body findings (outside the diff):** 0
- **Review id:** `4659798341`
- **Submit at:** <https://github.com/Khelsutra/badminton-highlight-indexer/pull/547/files>

Scroll to **Finish your review** ? click **Submit review**. Until you submit, the comments are visible only to you.

29. user

<user_query>

can you take another pass since the author has addressed your feedback.

</user_query>

30. assistant

I'll re-review PR #547 against the author's latest changes and leave a new PENDING review if needed.

31. tool_result

1?You are a meticulous code reviewer. Review code and produce structured review notes in a Markdown file at the path given in the prompt.

Process:

1. Read all relevant code thoroughly
2. Write findings to the specified review notes file
3. Use structured format: severity, file:line, description, suggestion, status

Rules:

- 10?- Check correctness first, style second
- Look for edge cases, error handling gaps, race conditions
 - Flag `unwrap()`, unnecessary `clone()`, or lock usage
 - Be specific: cite file:line for every issue
 - Do NOT fix the code yourself
 - In your final response, state the file path and summarize the verdict

32. tool_result

```

exit: 0
12e06e9f
github.com
  ? Logged in to github.com account avidullu (/home/avidullu/.config/gh/hosts.yml)
  - Active account: true
  - Git operations protocol: https
  - Token: gho_*****
{"additions":1735,"baseRefName":"master","body":"## Summary\n\nTwo things, both about **release confidence**:\n\n1. **New ? `scripts/gates/perf/`:** a standalone, release-gating performance tool that compares the binary **deployed on `api.khelsutra.guru`** against **repo HEAD**, on the local GPU, with no SPA / no user step. It times the four release-critical stages independently ? **checksum ? validation ? segmentation (`wasb_hybrid`, GPU via WSL) ? full-reel stitch** ? and gates the release (exit 1 on a stitch/total regression beyond tolerance).\n2. **Reorg ? consolidate the scattered nightly CLIs into `scripts/gates/`:** a cohesive gate package (quality + performance legs) instead of loose files, honoring the `docs/CODE_MAP.md` \"standalone CLIs live in `scripts/`\" convention.\n\n## The perf tool\n\n- **Prod resolution:** live Cloud Run image digest ? `perf_manifest.json` digest?commit map ? pinned ? `--prod-commit`. (Confirmed live: `sha256:7b62a8df? ? 304af7b`.)\n- **Fair comparison:** each commit runs from its own throwaway `git worktree`, one canonical config, identical inputs ? only the *code* differs. `skip_ai_handoff=true` + fast `stream_video` decode = deterministic + cloud-free. Always stitches the entire reel (watermark on, libx264 both sides).\n- **Self-help preflight** names each missing prereq + its fix. Flags: `--preflight`, `--stitch-only`, `--profile quick|full|stress`, `--update-baseline`.\n- **Exit codes** match the quality gates: `0 PASS` · 1 REGRESSION · 2 nothing-ran/preflight · 4 STALE.\n- Videos referenced by **basename + `--vault-dir`** (default `~/Videos`) so no personal path is committed (leak-scan guard).\n\n### Nightly integration\n\n`scripts/gates/nightly_perf.ps1` mirrors `nightly_regression.ps1`;\n`register_nightly_task.ps1` now installs **both** `KhelSutraNightlyRegression` (quality, 30-min cap) and `KhelSutraNightlyPerf` (perf, 3-hr cap ? a separate task since a GPU run far exceeds the quality cap).\n\n## The reorg\n\n`scripts/nightly_regression.py` / `nightly_reelcount.py` / `nightly_email.py` ?\n`scripts/gates/quality/`nscrip`nightly_regression.ps1` / `register_nightly_task.ps1`\n? `scripts/gates/`n+ `scripts/gates/perf/` (new) + `scripts/gates/{README.md, nightly_perf.ps1}`\n\n`moves` preserve git history. **Every** reference updated in-tree: the deployed GCP nightly (`run_on_vm.sh` import + invocation), `tests/test_nightly_` loaders, `eval_baselines/` docstrings, and the active docs (`CODE_MAP`, `NIGHTLY_ENGINE_REGRESSION`, `QUALITY_ITERATIONS`, `docs/README`, `ALPHA_LAUNCH_READINESS`, `DECISION_SNAPSHOT_REGRESSION`). Historical `docs/archives/` left as-is.\n\n## What the tool found (this run ? RTX 2070, prod `304af7b` vs HEAD `3249451`, 3 clips)\n\n**Performance-neutral, no regression.**\nchecksum/validation/segmentation unchanged (same detector/inputs). The one code-level delta ? `#531` single-pass stitch ? lands **within run-to-run noise**: matrix stitch ? was ?10% / ?13% / +5% across clips, and the controlled A/B (median of reps) was 1080p **+6.5%** vs 5.3K **?9.8%** (directions disagree). `#531`'s real win is the L4 **NVENC** path, not reproducible with local libx264. The other 8 release commits are cloud-only.\n\n## Verification\n\n- `ruff check` (pinned **0.15.16**) ? · `mypy` backend training ? · full suite **1933 passed / coverage 84.28%** (floor 70) ?\n- Tool **self-tested end-to-end** (preflight 9/9, prod auto-resolve, worktrees, stitch A/B, gating, report, exit 0). The self-test caught + fixed two Windows bugs in the tool: `gcloud` .CMD` resolution, and worktree `MAX_PATH` (added `core.longpaths=true` + a temp worktree root).\n- ?? 2 pre-existing `test_storage.py` failures (GCSFuse `#531`/`#535`) are **Windows path-separator** issues unrelated to this change ? they pass on CI Linux (master is green). Filed separately.\n\n## Caveats\n\n- Local, in-process measurement ? cloud-topology wins in the release (GCSFuse mount `#535`, CP `acquire_local_input` gating `#539`, cancel `#538`) can't be exercised locally.\n- 8 GB RTX 2070 vs prod's 24 GB L4: absolute segmentation times aren't comparable to prod; the prod-vs-HEAD *delta on one box* is the signal.\n- `--update-baseline` seeds `eval_baselines/perf_baseline.json` on the owner box (machine-specific, like the quality baselines ? not committed).\n\n? Generated with [Claude Code](https://claude.com/claude-code)\n\n", "comments": [{"id": "IC_kwDOSg1-f8AAAABJVvyHg", "author": {"login": "avidullu"}, "authorAssociation": "MEMBER", "body": "Thanks for the thorough review ? all **4 bugs, 7 suggestions, 2 nits** addressed in `8030dac`, plus the requested **Linux support**.\n\n### Bugs\n| finding | fix |\n|---|---|\n| `ps1` default `$RepoRoot` walked up 2 dirs from `scripts/gates/` ? `scripts/`; tasks armed with `scripts\\scripts\\gates\\` and permanently SKIPPED | **3 `Split-Path` hops** in all three wrappers (`register_nightly_task.ps1`, `nightly_perf.ps1`, `nightly_regression.ps1`) | gate passed a broken HEAD (missing gated stage ? `pct` None ? no finding ? exit 0) | a gated stage **prod produced but HEAD did not** (crash/timeout/0 segments) is now a **REGRESSION** |\n| silent PASS when nothing measurable ran (`--stitch-only` / missing sources / all-errors) |"}]}

```

```

`build()` returns `prod_observed`; `0` ? **exit 2** |\n\n### Suggestions\n| finding | fix
|\n|---|---|\n| gating `total_s` is unfair (segmentation noise dominates) | gate **stitch
only**; full-matrix `stitch_s` gated **only when prod/rc segment counts match** (else
`skipped`/\`not comparable\`). `gate.stages=[\"stitch_s\"]` |\n| exit-3 dead / inconsistent |
removed the wrapper's `3` branch + dropped \"3 no baseline\" from perf docs ? perf = **0/1/2/4**
(3 is quality-only) |\n| no unit tests for the gate core | **tests/test_perf_gate.py** (11
offline tests, by-path load like `test_nightly_regression.py`): pct edges,
regression/improvement, failed-HEAD, nothing-ran, segment-mismatch, total-not-gated, precedence
|\n| stale paths in `perf/README.m
d` | rewrote ? operator paths now `scripts/gates/?`, quality siblings `scripts/gates/quality/?`
|\n| old ps1 deleted without shim | documented **re-register after merge** in both READMEs |\n|
exit-2?0 masks lasting env breaks | documented **Task Scheduler success ? the gate signal** ?
`LATEST_STATUS.txt`/report is authoritative |\n| prod-then-rc order biases HEAD | **alternate
order per clip** (and per stitch-A/B case), logged |\n\n### Nits\n- `run.py` docstring
companions ? `scripts/gates/quality/?`. \n- harness `random.seed(0)` + the segment-count guard
above (fixed-pair stitch A/B is the apples-to-apples signal). \n\n### Linux support
(requested)\n`preflight` is **OS-aware** (WSL on Windows; native
`WASB_REPO_DIR`/\`WASB_WEIGHTS`/\`WASB_PYTHON` on Linux); harness selects `detector_impl=native`
off Windows (in-process wasb, like the GCP GPU-VM nightly). New README **$Running on Linux**.
**Verified in WSL** ? native checks fire. \n\n**Verification:** `ruff` 0.15.16 clean repo-wide .
full suite **1944 passed** / cov 84.28% . preflight **9/9** on Windows + native branch on Linux.
(The 2 `test_storage` GCSFuse failures are pre-existing **Windows-only**, unrelated ? fixed in #
548.)\n", "createdAt": "2026-07-
09T05:06:51Z", "includesCreatedEdit": false, "isMinimized": false, "minimizedReason": "", "reactionGrou
ps": [], "url": "https://github.com/Khelsutra/badminton-highlight-indexer/pull/547#issuecomment-
4921750046", "viewerDidAuthor": true}], "commits": [{"authoredDate": "2026-07-
09T04:20:27Z", "authors": [{"email": "avi.dullu@gmail.com", "id": "MDQ6VXNlcjE2NzQ4Nzc=", "login": "avi
dullu", "name": "Avi
Dullu"}, {"email": "noreply@anthropic.com", "id": "MDQ6VXNlcjgxDQ3", "login": "claude", "name": "Claude
Opus 4.8"}], "committedDate": "2026-07-09T04:20:27Z", "messageBody": "?ts/gates/\n\nAdd
scripts/gates/perf/ ? a standalone, release-gating tool that times\nchecksum -> validation ->
segmentation -> full-reel-stitch for the commit\ndeployed on api.khelsutra.guru vs repo HEAD, on
the local GPU (wasb_hybrid via\nWS
L), and gates the release (exit 1 on a stitch/total regression beyond\ntolerance). Preflight
self-help, prod auto-resolve from the live Cloud Run\nimage, isolated git worktrees, dated
reports under output/nightly_perf/. \n\nConsolidate the scattered nightly CLIs into a cohesive
scripts/gates/ package\n(honoring the docs/CODE_MAP.md \"standalone CLIs live in scripts/\")
convention):\n nightly_regression.py / nightly_reelcount.py / nightly_email.py ->
gates/quality/\n nightly_regression.ps1 -> gates/ ; register_nightly_task.ps1 -> gates/ (now\n
registers BOTH the quality and the new performance scheduled tasks). \n\nAll references updated
in-tree: the deployed GCP nightly (run_on_vm.sh import +\ninvoication), tests, eval-baseline
docstrings, and the active docs (CODE_MAP, \nNIGHTLY_ENGINE_REGRESSION, QUALITY_ITERATIONS,
README, ...). Archives left as-is. \n\nVerified: ruff 0.15.16 clean, mypy clean, full suite 1933
passed / coverage\n84.28%. (2 pre-existing Windows-only test_storage GCSFuse #535 failures
are\nunrelated to this change and pass on CI Linux.) Tool self-tested end-to-end. \n\nCo-
Authored-By: Claude Opus 4.8 <noreply@anthropic.com>, "messageHeadline": "feat(gates): local
release perf gate + consolidate nightly into scrip?", "oid": "0aa6d4b06a2cd82be07c25205df30109e8fc
a61e"}, {"authoredDate": "2026-07-
09T05:05:48Z", "authors": [{"email": "avi.dullu@gmail.com", "id": "MDQ6VXNlcjE2NzQ4Nzc=", "login": "avi
dullu", "name": "Avi
Dullu"}, {"email": "noreply@anthropic.com", "id": "MDQ6VXNlcjgxDQ3", "login": "claude", "name": "Claude
Opus 4.8"}], "committedDate": "2026-07-09T05:05:48Z", "messageBody": "?nux support\n\nBugs:\n- [1]
ps1 default $RepoRoot walked up 2 dirs from scripts/gates/ (-> scripts/, not repo root), \n so
registered tasks got scripts\\scripts\\gates\\... paths and permanently SKIPPED. Now 3 hops in\n
register_nightly_task.ps1, nightly_perf.ps1, nightly_regression.ps1. \n- [2] gate passed a broken
HEAD: a gated stage prod produced but HEAD did not (crash/timeout/\n 0 segments
) is now a REGRESSION. \n- [3] silent PASS when nothing measurable ran (--stitch-only / missing
sources / all-errors) is\n now exit 2 (prod_observed == 0). \n\nSuggestions:\n- gate stitch only
(drop total_s: segmentation-noisy + dominates total); full-matrix stitch_s is\n gated only when
prod/rc segment counts match (else reported \"not comparable\"). \n- drop the dead exit-3 branch
from nightly_perf.ps1 + the shared exit tables (perf uses 0/1/2/4; \n 3 is quality-only). \n- add
tests/test_perf_gate.py (pct edges, regression/improvement, failed-HEAD, nothing-ran, \n
segment-mismatch, total-not-gated, exit precedence). \n- fix pre-move paths in
scripts/gates/perf/README.md; document re-register-after-merge and that\n LATEST_STATUS.txt
(not Task Scheduler success) is the gate signal. \n- alternate prod/rc run order per clip to

```

cancel first-vs-second-run bias.\n\nNits:\n- run.py docstring companion paths ->
scripts/gates/quality/*; harness random.seed(0).\n\nLinux support:\n- preflight is OS-aware: WSL
checks on Windows, native wasb env (WASB_REPO_DIR / WASB_WEIGHTS /\n WASB_PYTHON) on Linux;
harness selects detector_impl=native off Windows (in-process wasb, like\n the GCP GPU-VM
nightly). README gains a \"Running on Linux\" section. Verified in WSL.\n\nVerified: ruff
0.15.16 clean repo-wide, full suite 1944 passed / coverage 84.28% (the 2\ntest_storage GCSFuse
#535 failures are pre-existing Windows-only, unrelated, fixed in PR #548),\npreflight 9/9
Windows + native branch in WSL.\n\nCo-Authored-By: Claude Opus 4.8
<noreply@anthropic.com>","messageHeadline":"fix(perf-gate): address review (4 bugs + 7
suggestions + 2 nits) + Li?","oid":"8030dacc01b0fa80e427898f2d2155a9ffc9b2bf"}], "deletions":114,
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UNCH_READINESS.md","additions":4,"deletions":4}, {"path":"docs/CODE_MAP.md","additions":5
,"deletions":5}, {"path":"docs/DECISION_SNAPSHOT_REGRESSION.md","additions":4,"deletions":4}, {"pa
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TIONS.md","additions":2,"deletions":2}, {"path":"docs/README.md","additions":1,"deletions":1}, {"p
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/perf_manifest.json","additions":40,"deletions":0}, {"path":"eval_baselines/reelcount_manifest.js
on","additions":1,"deletions":1}, {"path":"scripts/gates/README.md","additions":81,"deletions":0}
, {"path":"scripts/gates/nightly_perf.ps1","additions":83,"deletions":0}, {"path":"scripts/gates/n
ightly_regression.ps1","additions":3,"deletions":3}, {"path":"scripts/gates/perf/README.md","addi
tions":178,"deletions":0}, {"path":"scripts/gates/perf/config/bench_config.json","additions":64,"
deletions":0}, {"path":"scripts/gates/perf/harness.py","additions":341,"deletions":0}, {"path":"sc
ripts/gates/perf/preflight.py","additions":136,"deletions":0}, {"path":"scripts/gates/perf/run.py
","additions":453,"deletions":0}, {"path":"scripts/gates/perf/stitch_ab.py","additions":94,"delet
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quality/nightly_regression.py","additions":1,"deletions":1}, {"path":"scripts/gates/register_nigh
tly_task.ps1","additions":98,"deletions":0}, {"path":"scripts/register_nightly_task.ps1","additio
ns":0,"deletions":70}, {"path":"tests/test_nightly_reelcount.py","additions":2,"deletions":2}, {"p
ath":"tests/test_nightly_regression.py","additions":3,"deletions":1}, {"path":"tests/test_perf_ga
te.py","additions":121,"deletions":0}], "headRefName":"feat/perf-release-gate", "headRefOid":"8030
dacc01b0fa80e427898f2d2155a9ffc9b2bf", "number":547, "reviews":[{"id":"PRR_kwDOSg1-
f88AAAABFb7hRQ", "author":{"login":"avidullu"}, "authorAssociation":"MEMBER", "body":"## Summary\n
\nPR #547 consolidates the nightly quality CLIs under `scripts/gates/quality/`, wires GCP
`run\_on\_vm.sh` + tests to the new paths, and adds a solid local release **performance** harness
(`scripts/gates/perf/`) that compares a resolved prod commit vs HEAD via git worktrees, shared
`bench\_config.json`, `skip\_ai\_handoff`, and stage timing. The reorg of quality scripts
(REPO_ROOT depth, import path, test loaders, GCP runner) looks coherent. Dominant risks:
broken default `\$RepoRoot` in the new PowerShell wrappers/register script (one `Split-Path`
short after the move), **the perf gate treating hard failures as non-regressions** (missing
stages ? `pct` returns `None` ? exit 0), and **silent PASS when nothing measurable ran** (empty
stitch/vault). Also: no pure-unit tests for the new gate logic, stale paths in `perf/README.md`,
and existing Windows scheduled tasks will no-op after delete-without-shim until re-
registered.\n\n## Issue counts by severity\n\n- bugs: 4\n- suggestions: 7\n- nits: 2\n","submitt
edAt":null, "includesCreatedEdit":false, "reactionGroups":[], "state":"PENDING", "commit":{"oid":"0a
a6d4b06a2cd82be07c25205df30109e8fca61e"}}, {"state":"OPEN", "title":"feat(gates): local release
perf gate + consolidate nightly into
scripts/gates/", "url":"https://github.com/Khelsutra/badminton-highlight-indexer/pull/547"}]

33. assistant

New commit `8030dac` addresses the prior feedback. Collecting the updated diff and launching a second-pass review.

34. tool_result

```
exit: 0
{"base_sha":"3249451d5c408f8e20bff03e5d4e4c07f0bd0d4d","head_sha":"8030dacc01b0fa80e427898f2d2155a9ffc9b2bf"}
number: 547
title: feat(gates): local release perf gate + consolidate nightly into scripts/gates/
url: https://github.com/Khelsutra/badminton-highlight-indexer/pull/547
head: 8030dacc01b0fa80e427898f2d2155a9ffc9b2bf
baseRef: master
```


files: 28

- deploy/gcp/nightly_regression/README.md
- deploy/gcp/nightly_regression/run_on_vm.sh
- docs/ALPHA_LAUNCH_READINESS.md
- docs/CODE_MAP.md
- docs/DECISION_SNAPSHOT_REGRESSION.md
- docs/NIGHTLY_ENGINE_REGRESSION.md
- docs/QUALITY_ITERATIONS.md
- docs/README.md
- eval_baselines/nightly_baseline.json
- eval_baselines/perf_manifest.json
- eval_baselines/reelcount_manifest.json
- scripts/gates/README.md
- scripts/gates/nightly_perf.ps1
- scripts/gates/nightly_regression.ps1
- scripts/gates/perf/README.md
- scripts/gates/perf/config/bench_config.json
- scripts/gates/perf/harness.py
- scripts/gates/perf/preflight.py
- scripts/gates/perf/run.py
- scripts/gates/perf/stitch_ab.py
- scripts/gates/quality/nightly_email.py
- scripts/gates/quality/nightly_reelcount.py
- scripts/gates/quality/nightly_regression.py
- scripts/gates/register_nightly_task.ps1
- scripts/register_nightly_task.ps1
- tests/test_nightly_reelcount.py
- tests/test_nightly_regression.py
- tests/test_perf_gate.py

diff_bytes: 138415

diff_lines: 2260

fix(perf-gate): address review (4 bugs + 7 suggestions + 2 nits) + Linux support

Bugs:

- [1] ps1 default \$RepoRoot walked up 2 dirs from scripts/gates/ (-> scripts/, not repo root), so registered tasks got scripts\scripts\gates\... paths and permanently SKIPPED. Now 3 hops in register_nightly_task.ps1, nightly_perf.ps1, nightly_regression.ps1.
- [2] gate passed a broken HEAD: a gated stage prod produced but HEAD did not (crash/timeout/0 segments) is now a REGRESSION.
- [3] silent PASS when nothing measurable ran (--stitch-only / missing sources / all-errors) is now exit 2 (prod_observed == 0).

Suggestions:

- gate stitch only (drop total_s: segmentation-noisy + dominates total); full-matrix stitch_s is gated only when prod/rc segment counts match (else reported "not comparable").
- drop the dead exit-3 branch from nightly_perf.ps1 + the shared exit tables (perf uses 0/1/2/4; 3 is quality-only).
- add tests/test_perf_gate.py (pct edges, regression/improvement, failed-HEAD, nothing-ran, segment-mismatch, total-not-gated, exit precedence).
- fix pre-move paths in scripts/gates/perf/README.md; document re-register-after-merge and that LATEST_STATUS.txt (not Task Scheduler success) is the gate signal.
- alternate prod/rc run order per clip to cancel first-vs-second-run bias.

Nits:

- run.py docstring companion paths -> scripts/gates/quality/*; harness random.seed(0).

Linux support:

- preflight is OS-aware: WSL checks on Windows, native wasb env (WASB_REPO_DIR / WASB_WEIGHTS / WASB_PYTHON) on Linux; harness selects detector_impl=native off Windows (in-process wasb, like the GCP GPU-VM nightly). README gains a "Running on Linux" section. Verified in WSL.

Verified: ruff 0.15.16 clean repo-wide, full suite 1944 passed / coverage 84.28% (the 2 test_storage GCSFuse #535 failures are pre-existing Windows-only, unrelated, fixed in PR #548), preflight 9/9 Windows + native branch in WSL.

35. tool_result

Subagent started in background.

subagent_id: 019f4547-7f0b-7423-8442-403d998bbdac

type: general-purpose

description: [reviewer] pr #547 pass 2

Use get_command_or_subagent_output with task_ids=["019f4547-7f0b-7423-8442-403d998bbdac"] and timeout_ms to wait for results.

36. tool_result

=== Task 019f4547-7f0b-7423-8442-403d998bbdac ===

Command: [subagent:general-purpose] [reviewer] pr #547 pass 2

Status: completed

Started: 2026-07-09T05:09:20Z

Ended: 2026-07-09T05:12:19Z

Duration: 179.71s

Exit Code: 0

Output File:

=== Output ===

Review complete

****Notes written to:**** ["/tmp/grok-review-12e06e9f.md"](/tmp/grok-review-12e06e9f.md)

Verdict

****Prior findings (1?12) are fixed correctly**** ? RepoRoot 3-hop, failed-HEAD REGRESSION for missing values, exit 2 when nothing measurable ran, stitch-only gating + segment-count skip, no exit 3, tests, docs/paths, re-register + Task Scheduler caveats, alternate order, `random.seed(0)`, and Linux preflight/`detector\_impl=native`.

****No merge blockers.**** Residual items are small follow-ups:

| Sev | Issue | |
|------------|--|--|
| ----- | ----- | |
| suggestion | Timed stitch **exceptions** still record numeric `stitch_s` ? can look like improvement (stitch A/B usually still catches by default) | |
| suggestion | `nightly_perf.ps1` maps exit 2 ? SKIPPED with a **preflight-only** message (also covers empty vault / nothing-ran) | |
| nit | Description still says **stitch/total** | |
| nit | Report hardcodes **WSL2 GPU** on Linux | |

<subagent_meta>id=019f4547-7f0b-7423-8442-403d998bbdac, type=general-purpose, tool_calls=26, turns=1, duration_ms=179707</subagent_meta>

<subagent_result>

subagent_id: 019f4547-7f0b-7423-8442-403d998bbdac

subagent_type: general-purpose

To continue this subagent's conversation, use

resume_from="019f4547-7f0b-7423-8442-403d998bbdac".

</subagent_result>

37. assistant

Prior findings are resolved. Processing residual issues and posting a second PENDING review.

38. tool_result

1?## Summary

Second-pass review of PR #547 (``8030dac``) confirms the author fixed the prior findings correctly: 3-hop ``$RepoRoot`` in all three `.ps1`` wrappers/register scripts, REGRESSION when prod has a gated stage and HEAD does not, exit 2 when ``prod_observed == 0``, stitch-only gating with segment-count skip, dropped exit-3, unit tests, path/docs updates, re-register + Task Scheduler caveats, alternate prod/rc order (matrix + stitch A/B), ``random.seed(0)``, and OS-aware Linux preflight/``detector_impl=native``. Dominant residual risk is a small incomplete edge of the failed-HEAD fix (exception path still records a numeric ``stitch_s``), plus a couple of doc/status-message nits from the Linux + gating retune. No blockers for merge from this pass; residual items are small follow-ups.

Issues

Issue 1 -- Severity: suggestion

- File: `scripts/gates/perf/harness.py:312`
- Description: When full-matrix stitch raises, the harness still records a numeric ``stages["stitch_s"]`` (wall time until exception) and only puts the failure in ``errors["stitch"]``. ``build()`` treats any numeric HEAD value as a successful measurement, so a fast crash can look like a large "improvement" rather than a failed gated stage. Default stitch A/B usually still catches a broken stitch path (exception ? no ``result.json`` ? no median ? REGRESSION), so this is mostly a hole for ``--no-stitch-ab`` / matrix-only runs and for confusing dual signals in the report. The prior "failed HEAD is REGRESSION" fix only covers ``None``/missing stage values, not timed exceptions.
- 10?- Suggestion: On stitch exception set ``stages["stitch_s"] = None`` (keep the error), and/or in ``build()`` treat presence of ``errors["stitch"]`` (or missing reel output) as "HEAD produced no value". Add a unit test: prod ``stitch_s=10``, rc ``stitch_s=0.1`` with ``errors={"stitch": "..."}`` ? regression.
- Status: open

Issue 2 -- Severity: suggestion

- File: `scripts/gates/nightly_perf.ps1:78`
- Description: Exit code 2 from ``run.py`` now means both preflight failure ``and`` "nothing measurable ran" (empty vault / all stitch sources missing / prod-side no gated observations). The wrapper always writes ``SKIPPED`` with detail ``environment not ready (preflight)`` and exits 0. Operators watching ``LATEST_STATUS.txt`` will mis-diagnose a missing vault/profile as a WASB preflight problem.
- Suggestion: Branch the status text on log content or pass a richer signal (e.g. distinct exit codes, or parse the last log line). At minimum, change the detail to something like ``exit 2`` (preflight failed or nothing measurable ran) ? see log.
- Status: open

Issue 3 -- Severity: nit

- 20?- File: `scripts/gates/nightly_perf.ps1:8`
- Description: Description still says the gate fires on a "stitch/total regression", but ``total_s`` was correctly dropped from ``gate.stages`` (manifest + ``build()`` only gate ``stitch_s``).
- Suggestion: Replace with "stitch regression" (or "stitch / stitch-A/B regression beyond tolerance").
- Status: open

Issue 4 -- Severity: nit

- File: `scripts/gates/perf/run.py:248`
 - Description: The markdown report hardcodes ``segmenter: wasb_hybrid (WSL2 GPU)`` even though Linux now runs ``detector_impl=native`` in-process. Misleading after the new Linux path.
 - Suggestion: Pick the label from ``sys.platform`` (or record ``detector_impl`` in ``meta`` during the run) so the report says WSL2 GPU on Windows and native GPU on Linux.
 - Status: open
- 30?

39. tool_result

exit: 0
OK: `scripts/gates/perf/harness.py:312` [suggestion]
OK: `scripts/gates/nightly_perf.ps1:78` [suggestion]
OK: `scripts/gates/nightly_perf.ps1:8` [nit]
OK: `scripts/gates/perf/run.py:248` [nit]

```
inline=4 promoted=0
wrote /tmp/grok-review-pending-12e06e9f.json (3833 bytes), comments=4
exit=1
gh: Unprocessable Entity (HTTP 422)
id= None
state= None
html_url= None
```

40. tool_result

```
exit: 0
gh: Unprocessable Entity (HTTP 422)
---
HTTP/2.0 422 Unprocessable Entity
Access-Control-Allow-Origin: *
Access-Control-Expose-Headers: ETag, Link, Location, Retry-After, X-GitHub-OTP, X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Used, X-RateLimit-Resource, X-RateLimit-Reset, X-OAuth-Scopes, X-Accepted-OAuth-Scopes, X-Poll-Interval, X-GitHub-Media-Type, X-GitHub-SSO, X-GitHub-Request-Id, Deprecation, Sunset, Warning
Content-Length: 217
Content-Security-Policy: default-src 'none'
Content-Type: application/json; charset=utf-8
Date: Thu, 09 Jul 2026 05:12:48 GMT
Referrer-Policy: origin-when-cross-origin, strict-origin-when-cross-origin
Server: github.com
Strict-Transport-Security: max-age=31536000; includeSubdomains; preload
Vary: Accept-Encoding, Accept, X-Requested-With
X-Accepted-OAuth-Scopes:
X-Content-Type-Options: nosniff
X-Frame-Options: deny
X-Github-API-Version-Selected: 2022-11-28
X-Github-Media-Type: github.v3; format=json
X-Github-Request-Id: 35E1:AC094:A8CC41:BBA3E3:6A4F2DCF
X-Oauth-Client-Id: 178c6fc778ccc68e1d6a
X-Oauth-Scopes: gist, read:org, repo, workflow
X-Ratelimit-Limit: 5000
X-Ratelimit-Remaining: 4998
X-Ratelimit-Reset: 1783577562
X-Ratelimit-Resource: core
X-Ratelimit-Used: 2
X-Xss-Protection: 0

{"message":"Unprocessable Entity","errors":["User can only have one pending review per pull request"],"documentation_url":"https://docs.github.com/rest/pulls/reviews#create-a-review-for-a-pull-request","status":"422"}gh: Unprocessable Entity (HTTP 422)
commit_id: 8030dacc01b0fa80e427898f2d2155a9ffc9b2bf
scripts/gates/perf/harness.py 312 RIGHT body_len 971
scripts/gates/nightly_perf.ps1 78 RIGHT body_len 639
scripts/gates/nightly_perf.ps1 8 RIGHT body_len 285
scripts/gates/perf/run.py 248 RIGHT body_len 341
body preview: ## Summary

Second-pass review of PR #547 (`8030dac`) confirms the author fixed the prior findings correctly: 3-hop `$RepoRoot` in all three `.ps1` wrappers/register scripts, REGRESSION when prod has
{"commit_id":"0aa6d4b06a2cd82be07c25205df30109e8fca61e","id":4659798341,"state":"PENDING","user":"avidullu"}
```

41. assistant

An earlier PENDING review is blocking a new one. Deleting the stale first-pass review, then posting the second-pass findings.

42. tool_result

```

exit: 0
{"id":4659798341,"node_id":"PRR_kwDOSg1-
f88AAAABFb7hRQ","user":{"login":"avidullu","id":1674877,"node_id":"MDQ6VXNlcjE2NzQ4Nzc=", "avatar
_url":"https://avatars.githubusercontent.com/u/1674877?v=4","gravatar_id":"","url":"https://api.
github.com/users/avidullu","html_url":"https://github.com/avidullu","followers_url":"https://api.
github.com/users/avidullu/followers","following_url":"https://api.github.com/users/avidullu/fol
lowing/{other_user}","gists_url":"https://api.github.com/users/avidullu/gists/{gist_id}","starre
d_url":"https://api.github.com/users/avidullu/starred{/owner}/{repo}","subscriptions_url":"https
://api.github.com/users/avidullu/subscriptions","organizations_url":"https://api.github.com/user
s/avidullu/orgs","repos_url":"https://api.github.com/users/avidullu/repos","events_url":"https:/
/api.github.com/users/avidullu/events{/privacy}","received_events_url":"https://api.github.com/u
sers/avidullu/received_events","type":"User","user_view_type":"public","site_admin":false},"body
":"## Summary\n\nPR #547 consolidates the nightly quality CLIs under `scripts/gates/quality/`,
wires GCP `run_on_vm.sh` + tests to the new paths, and adds a solid local release
**performance** harness (`scripts/gates/perf/`) that compares a resolved prod commit vs HEAD via
git worktrees, shared `bench_config.json`, `skip_ai_handoff`, and stage timing. The reorg of
quality scripts (REPO_ROOT depth, import path, test loaders, GCP runner) looks coherent.
Dominant risks: **broken default `$RepoRoot` in the new PowerShell wrappers/register script**
(one `Split-Path` short after the move), **the perf gate treating hard failures as non-
regressions** (missing stages ? `pct` returns `None` ? exit 0), and **silent PASS when nothing
measurable ran** (empty stitch/vault). Also: no pure-unit tests for the new gate logic, stale
paths in `perf/README.md`, and existing Windows scheduled tasks will no-op after delete-without-
shim until re-registered.\n\n## Issue counts by severity\n\n- bugs: 4\n- sugge
stions: 7\n- nits: 2\n","commit_id":"0aa6d4b06a2cd82be07c25205df30109e8fca61e","state":"PENDING"
,"html_url":"https://github.com/Khelsutra/badminton-highlight-indexer/pull/547#pullrequestreview
-4659798341","pull_request_url":"https://api.github.com/repos/Khelsutra/badminton-highlight-
indexer/pulls/547","_links":{"html":{"href":"https://github.com/Khelsutra/badminton-highlight-
indexer/pull/547#pullrequestreview-4659798341"},"pull_request":{"href":"https://api.github.com/repos/Khelsutra/badminton-highlight-
indexer/pulls/547"}}, "author_association":"MEMBER"}delete_exit=0
post_exit=0
id= 4660014889
state= PENDING
html_url= https://github.com/Khelsutra/badminton-highlight-
indexer/pull/547#pullrequestreview-4660014889

```

43. tool_result

```

exit: 0
cleaned

```

44. assistant

Second-pass review complete ? PR #547

```

**PR:** [feat(gates): local release perf gate + consolidate
nightly?](https://github.com/Khelsutra/badminton-highlight-indexer/pull/547)
**Head reviewed:** `8030dac` (review-response + Linux support)

```

Prior findings status

All **13** first-pass items (4 bugs + 7 suggestions + 2 nits) are **verified fixed**:

- 3-hop `\$RepoRoot` in all three `.ps1` files
- REGRESSION when prod has a gated stage and HEAD does not
- Exit 2 when nothing measurable ran
- Stitch-only gating + segment-count skip
- Exit 3 dropped for perf
- `tests/test\_perf\_gate.py`
- Doc/path updates, re-register + Task Scheduler caveats
- Alternating prod/rc order, `random.seed(0)`, Linux preflight/`native`

Residual issues (this pass)

| Severity | Count | Notes |
|----------------|-------|---|
| **bug** | 0 | No merge blockers |
| **suggestion** | 2 | Timed stitch exceptions still record numeric `stitch_s`; exit-2 status text always says ?preflight? |
| **nit** | 2 | ?stitch/total? wording; report hardcodes ?WSL2 GPU? on Linux |

PENDING review posted

- **Inline comments:** 4
- **Body findings (outside the diff):** 0
- **Review id:** `4660014889`
- **Submit at:** <https://github.com/Khelsutra/badminton-highlight-indexer/pull/547/files>

Note: the first-pass PENDING review was deleted so this second pass could be posted (GitHub allows only one pending review per user per PR). Until you submit, comments are visible only to you.

Verdict: Prior feedback fully addressed; residuals are small follow-ups, not blockers.